

**CITYINSIGHT: WEB AND MOBILE-BASED PERSONAL SAFETY
COMPANION**

A Capstone Project
Presented to the Faculty of the
Information Technology Department
Pangasinan State University – Urdaneta City Campus
Urdaneta City, Pangasinan

In Fulfillment of the Requirements for the Course Capstone 2
for the Bachelor of Science in Information Technology

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December 2024

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude to all those who have contributed to the completion of this research project.

First and foremost, we extend our heartfelt thanks to Mr. Nahum Quiros for their invaluable guidance, support, and encouragement throughout the research process. Their expertise and insightful feedback have been instrumental in shaping this study.

We also wish to thank the participants who generously shared their time and insights, without whom this research would not have been possible.

Furthermore, we are grateful to Pangasinan State University for providing access to resources and facilities that facilitated our research efforts.

Finally, we would like to acknowledge the unwavering support of our families and friends, who stood by us with patience and encouragement during this endeavor.

Thank you all for your contributions and support.

DEDICATION

We dedicate this project to the people of Urdaneta City, especially its residents, students, and local authorities specifically the Urdaneta City Police and CDRMO who deal with daily safety and navigation issues. We are deeply grateful to our families for their support, patience, and encouragement throughout this journey. We also thank our friends, mentors, and respondents who gave their time and ideas to help us improve CityInsight.

We hope future users of CityInsight will find it helpful for staying safe and navigating better. This project is for you and the goal of a safer and better-informed community. May this inspire more ideas and teamwork to solve urban challenges.

ABSTRACT

Crime and flooding are common problems in Urdaneta City that affect public safety and navigation. This study introduces CityInsight, a web and mobile-based app designed to provide real-time alerts on hazards and emergency services. With features like crime policies and flood-prone area updates, CityInsight helps users make better decisions and avoid unsafe areas, making them feel more secure while exploring new places. The research used SUMI methods to make sure the app was developed efficiently and focused on user needs. Data were gathered to find out what features were needed and to ensure the app met the needs of Urdaneta City's people. The system was made to be easy to use, reliable, and deliver real-time updates, with user feedback helping to improve it. Testing showed that CityInsight makes safety, navigation, and decision-making easier in emergencies. The study shows how technology can help solve safety problems in cities and gives a guide for creating similar solutions in other places. CityInsight aims to make the community safer and more confident, opening doors for better tools in personal safety and navigation.

Keywords: Real-Time Safety Updates, Urban Navigation, Personal Companion, Emergency Services, Urdaneta City

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CHAPTER I

INTRODUCTION

Situation Analysis

The inter-play between human activities and natural events manifests specifically as man-made hazards. Crime is described as a hazard because of its uncertain probability for most households, communities, and organizations as it occurs when a community, organization or households suffers from shocks or stresses of floods, wars and terrorism which not only overwhelms their capacity to cope but also respond and recover. This also prevents them from attaining sustainable development. Kwanga, Shabu, Adaaku, (2017).

Various crime mapping systems and technologies have been implemented to bolster public safety and improve law enforcement effectiveness. One notable system is the Crime Incident Reporting System (CIRS), developed by the Philippine National Police (PNP). CIRS allows for real-time reporting and mapping of crime incidents, which helps track and analyze criminal activities across different regions. This system provides law enforcement agencies with valuable data to identify crime patterns and trends, facilitating more informed decision-making and resource allocation.

In accordance with this, the PNP has introduced the Crime Information Reporting and Analysis System (CIRAS), which consolidates crime data and includes crime mapping features. CIRAS enables the visualization of crime hotspots and supports strategic planning by analyzing crime trends over time. This system aids in developing targeted interventions and preventive measures to address crime more effectively.



Local government units (LGUs) in the Philippines also utilize crime mapping tools to enhance community safety. Cities such as Quezon City and Makati have implemented their own crime mapping systems, often incorporating Geographic Information System (GIS) technology. These local systems help monitor and address crime within specific jurisdictions by providing detailed maps and analysis of crime distribution.

The Philippines is one of the most disaster-prone countries in the world, primarily due to its geographic location in the Pacific Ring of Fire and its tropical climate. This positioning exposes the country to frequent seismic activities, including earthquakes and volcanic eruptions, as well as extreme weather events like typhoons, floods, and landslides. On average, the Philippines is hit by about 20 typhoons each year, with some of the most destructive in recent history being Typhoon Haiyan (Yolanda) in 2013 and Typhoon Rolly (Goni) in 2020, which caused massive loss of life, displacement of communities, and extensive damage to infrastructure (Lagmay et al., 2017). Additionally, the country has experienced significant earthquakes, such as the 1990 Luzon earthquake, which resulted in severe casualties and destruction, and the eruption of Mount Pinatubo in 1991, one of the largest volcanic eruptions of the 20th century, affecting global climate and displacing thousands of people (Bankoff, 2003). The archipelagic nature of the Philippines, combined with high population density, deforestation, and poor urban planning, exacerbates the risks of natural disasters. Many areas are isolated and difficult to reach, complicating disaster preparedness, response, and recovery efforts. This constant exposure to natural hazards poses significant challenges to sustainable development and poverty reduction in the country (Heijmans, 2001).



Additionally, heavy rains and typhoons in the Philippines frequently cause extensive damage to land and structures, putting rural communities in critical conditions. A study by Gacu (2023) reveals that between 2008 and 2013, the five most expensive typhoon events led to USD 138 million in damages, highlighting the need for better risk assessment tools and mitigation strategies. The study employed a structure vulnerability assessment using geographic information systems (GIS) and local flood hazard maps to create a risk map indicating building vulnerability. Out of 3,094 structures analyzed, 76.79% were classified as high risk, with 10.25% at moderate risk and 12.94% at very high risk. The resulting maps can help local governments prioritize flood risk areas, plan mitigation strategies, and educate communities on improving structural integrity to enhance resilience against flooding.

Geographic Information Systems (GIS) have become an essential tool for flood management in the Philippines, significantly enhancing the country's capacity to address flood risks. GIS technology is employed to create detailed flood hazard maps by integrating various data layers, such as topography, land use, and historical flood data, which helps identify flood-prone areas and assess potential impacts. This capability is crucial for local government units (LGUs) to prioritize flood mitigation measures and develop tailored strategies for their communities.

In addition, GIS integrates real-time data from automated weather stations, rain gauges, and flood monitoring systems, which supports early warning systems by providing up-to-date information on water levels and rainfall. This integration allows for timely flood alerts, improving preparedness and response efforts. During and after flood events, GIS aids in disaster response by analyzing satellite imagery and aerial surveys to assess damage, coordinate relief



efforts, and plan recovery operations. Furthermore, GIS supports public awareness and education by visualizing flood risks and potential impacts, helping communities better understand and prepare for flooding. Despite these advancements, challenges such as data accuracy, integration across various systems, and the need for continuous updates and maintenance remain significant. Addressing these challenges is crucial for maximizing the effectiveness of GIS in flood risk management and enhancing overall disaster resilience in the Philippines.

Urdaneta City, a significant urban center in Pangasinan, Philippines, is facing considerable challenges due to rapid urbanization. The city's growth has led to increased crime rates and heightened vulnerability to flooding, particularly during the rainy season. This urban expansion has resulted in a rise in criminal activities, from petty thefts to more serious offenses. Although the local police force is actively working to manage these issues, the increasing crime rates and the city's geographical susceptibility to flooding, exacerbated by the absence of an effective early warning system, present ongoing concerns.

The Philippine National Police (PNP) has implemented the Crime Incident Reporting System (CIRS) since 2016 to address crime through the mapping of incidents and identification of hotspots. This system allows law enforcement agencies to analyze crime patterns and deploy resources more effectively. However, localized tools are still needed to improve crime monitoring and response in Urdaneta City.



The city's flood management relies on community-based alert systems managed by Barangay Disaster Risk Reduction and Management Offices (BDRRMO). These systems use mobile phones, community radios, and loudspeakers to disseminate flood warnings and safety information. Despite these efforts, the effectiveness of flood risk management could be enhanced. Research supports the importance of localized data for flood management. Jongman, Ward, and Aerts (2018) emphasize that effective early warning systems are crucial for reducing flood exposure and enhancing resilience in flood-prone urban areas.

CityInsight plans to come up as a web and mobile application designed to provide insights into crime rates and safety through crowdsourced information. By leveraging this system, individuals can ease their uncertainties about crimes and natural disasters that may occur unexpectedly, as CityInsight aims to serve as their trusted companion, offering guidance and protection in navigating their surroundings.

Statement of Objectives

This research aims to design and develop the CityInsight: Web and Mobile-based Personal Companion. CityInsight will provide timely safety updates, such as alerts about hazards and emergency services, helping users make informed decisions and avoid risky areas. This real-time information will boost their confidence and security while exploring new places, ultimately improving their navigation experience.

Specifically:



1. To determine the policies and procedures needed in response to different crime incidents and obtain valuable insights regarding the flood prone areas in Urdaneta City.
2. To identify the needed features CityInsight: Web and Mobile-based Personal Companion.
3. To develop CityInsight: Web and Mobile-based Personal Companion using Scrum methodology.
4. To evaluate the usability of the system using Software Usability Measurement Inventory (SUMI) Questionnaire.

Significance of the Study

The possibility for this study to change urban safety measures through creative technological solutions makes it very significant. CityInsight can benefit multiple stakeholders by using crowdsourcing techniques to collect plenty of information on crime rates and natural disasters:

Local Residents. By giving people access to real-time safety information, CityInsight allows people to act for their own personal safety and make informed choices.

Communities. By raising awareness and preparing people better, CityInsight builds community security and allows them to deal with threats together.

Urban Planners and Policy Makers. The information provided by CityInsight allows well-informed decision-making about urban safety measures, targeted actions and efficient resource distribution.



Researchers. This research will help the enhancement of the researcher's ability, knowledge, and skills in software development.

PNP Emergency Responders. CityInsight enables PNP emergency responders to better understand emerging safety concerns and coordinate timely responses, ultimately saving lives.

Society as a Whole. The application will serve as an example for future creative solutions to improve capacities and overall wellness by demonstrating the possibilities of shared intelligence and technology for solving social issues.

Students. The development of the application supports academic creativity by demonstrating how cooperative intelligence and technology may improve understanding and address social problems, guiding the way for the future academics toward meaningful study.

Future Researchers. For up-and-coming researchers, the application provides a powerful example of how technology and collective intelligence may be used to address difficult issues in society, inspiring them to follow creative study guidance.

By responding to the needs of these individuals, this study improves academic understanding while also having real-world effects to improve safety and flexibility in urban environments.



Scope and Limitations of the Project

The scope of this study encompasses the development and implementation of CityInsight, a web and mobile application designed to serve as a safety companion for the residents of Urdaneta City. This application aims to enhance users' confidence in exploring and engaging with various barangays within Urdaneta City by providing features such as location tagging and notification of specific hazards or unsafe areas. CityInsight enables users to tag or ping locations that are prone to specific hazards or unsafety, thereby allowing other users to be notified when they enter these tagged locations. Additionally, users have the capability to tag events such as floods, earthquakes, and other emergencies in their respective areas. Furthermore, CityInsight offers a one tap emergency alert feature, allowing users to send alerts to the administrative side of the application. Upon receiving these alerts, administrators can take appropriate actions for the specific events reported by users. This may involve coordinating with specialized government units such as Philippine National Police (PNP), and City Disaster Risk Reduction Management Office (CDRRMO) to address the reported event effectively. The study is delimited to the development, implementation, and evaluation of CityInsight within the context of Urdaneta City. It does not extend to the development of physical infrastructure or specialized emergency response protocols outside the capabilities of the application. Additionally, the study does not encompass the broader socio-economic implications of safety measures or emergency response systems in Urdaneta City.

Definition of Terms

Alert Notification. This system continuously monitors the user's position in real-time, cross-referencing it with a database of known dangerous areas, such as high-crime zones and flood-



prone areas. Upon detecting proximity to one of these locations, the device instantly sends a notification that may include sound, vibration, and a visual message, informing the user of the danger and advising them on how to avoid or navigate it safely. This proactive approach helps ensure the user's safety by providing timely and actionable alerts whenever they are at risk.

CityInsight. A service or platform that offers data and resources about crime, safety, and neighborhood alerts in cities with the goal of improving individual safety.

Crime Data. Information on crimes reported in a specific area, often collected, and maintained by law enforcement agencies.

Crowdsourcing. Collect and share community insights to report and verify criminal activities or suspicious behavior, enhancing local safety through collective vigilance.

Data Integration. The process of combining data from multiple sources (e.g., crime data from law enforcement, historical data sets) into a single system like CityInsight.

Emergency Hotlines. Provides emergency hotlines, which are telephone lines established to provide immediate assistance, support, and intervention during various types of emergencies.

Geolocation Technology. Technology that uses location data to determine a user's physical location.

Geotagging. Emergency response feature that uses radius and range for precise real-time location pinpointing to speed up aid and guarantee efficient crisis management.

Incident History Logs. Review a comprehensive record of crime reports and alerts in your area, providing valuable insights into local safety trends and patterns.



Mobile-based. Applications and services created especially for use on portable electronics, such smartphones and tablets, frequently have features optimized for portability and access while on the go.

Personal Safety Companion. A tool or service designed to assist individuals in maintaining their safety by providing relevant information, guidance, and support tailored to their specific needs and situations.

Procedures. Step-by-step instructions on how to respond to crime incidents and collect important information about flood-prone areas in Urdaneta City. They guide what actions to take and how to do them properly, ensuring that everything is done efficiently and effectively.

Processes. Actions taken to respond to crime incidents and collect important information about flood-prone areas in Urdaneta City. These steps guide how tasks should be done, from reacting to an incident to gathering and analyzing data, ensuring everything is handled consistently and effectively.

Protocols. Step-by-step procedures that guide how to respond to crime incidents and collect important information about flood-prone areas in Urdaneta City. These protocols ensure that everyone involved knows what to do, who to contact, and how to gather and use information to keep the community safe and better prepared for future incidents.

Safety Rating. A score or indicator displayed from the application that reflects the safety of a user's current location. This rating is likely based on a combination of factors like crime statistics and other safety data.



Safety Tips. Discover practical advice and effective strategies for staying safe in different scenarios, helping you navigate potential risks with confidence.

Scalability. The ability of CityInsight to be adapted and expanded to cover new geographic locations or incorporate additional features in the future.

User-Friendly Interface. A design for the CityInsight app that is easy to navigate and understand, allowing users to access safety information intuitively.

Web-based. refers to applications and services that can only be accessed and used with an internet connection via a web browser on a computer or mobile device



CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presented a review of the related literature and studies for the capstone project entitled “CityInsight: Web and Mobile-based Safety Companion”. This chapter also presents the synthesis and project framework to fully understand the research for better comprehension of the study.

Procedures and Protocols of Crime Incident Response and Flood Management

The Philippine National Police (PNP) utilizes a range of procedures to effectively respond to crime incidents and ensure public safety. These procedures include immediate reporting of incidents through hotlines, mobile applications, or directly at police stations. The PNP follows standardized protocols for investigating crimes, which involve systematic evidence collection, witness interviews, and thorough case documentation. For emergency response, the PNP deploys specialized units, such as SWAT or anti-crime teams, to address urgent situations and coordinates closely with other law enforcement agencies and emergency services to manage and resolve incidents efficiently. Furthermore, the PNP maintains a centralized system for logging crime incidents and performs regular analyses of crime data to identify patterns, trends, and areas that require attention, which aids in improving overall crime management strategies.

In parallel, the National Disaster Risk Reduction and Management Council (NDRRMC) focuses on flood management and disaster preparedness. Their procedures include conducting comprehensive flood risk assessments by mapping flood-prone areas and evaluating the



potential impacts of floods on communities and infrastructure. The NDRRMC develops detailed flood response plans, which are regularly updated to address evolving risks, and employs early warning systems to alert communities and prepare them for potential flooding. Coordination with various agencies is crucial, as the NDRRMC works with local government units and emergency services to manage resources and support flood response efforts. This includes deploying rescue teams, medical aid, and relief supplies during flood emergencies to ensure effective disaster management and community support. By integrating these procedures, both the PNP and NDRRMC contribute significantly to enhancing public safety and resilience in the Philippines.

Spatial analysis of violent crimes in Metro Manila, Philippines

Mojica, V. J., Choi, A., Leong, R. N., Co, F. (2017). Spatial patterns of murder and physical injury in Metro Manila, Philippines were visualized through conditional choropleth maps. Relationship of both crime rates with some demographic variables were investigated while accounting for possible spatial autocorrelation using spatial lag models. Results show that both crime rates tend to cluster in the northern cities of Metro Manila. Furthermore, significant spatial lag coefficients were found only for physical injury rates, with values ranging from 0.49 to 0.62, signifying a positive city-level spatial dependence of physical injury rates in Metro Manila. Moreover, some demographic covariates, such as population density, percentage of young males, education, marriage, and immigration were found to be associated with both crime rates. These results could serve as useful indicators of crime incidence; thus, it is recommended that crime monitoring systems include them to aid in resource allocation and program planning for better crime prevention and security management. Mojica et al. (2017) analyzed murder and



physical injury spatial patterns in Metro Manila, finding clusters in the northern cities. They identified demographic factors like population density and education associated with crime rates. These insights could improve crime monitoring and aid in resource allocation for better crime prevention and security management. Such findings align with CITYINSIGHT's goal to provide comprehensive safety information, enhancing community safety by considering spatial and demographic factors.

Co-Design for Enhancing Flood Resilience in Davao City, Philippines

Miyamoto, M. et al. (2022). Enhancing flood resilience requires a multifaceted approach that combines structural measures with social capacity building and early warning systems. As climate change is expected to exacerbate flood risks in the future, addressing these challenges becomes increasingly crucial. Recent studies highlight the importance of integrating various strategies to effectively mitigate flood damage and improve community resilience.

One significant study focused on Davao City, Philippines, which aimed to develop a comprehensive methodology for enhancing flood resilience through improved disaster literacy and active stakeholder engagement. This study introduced the Online Synthesis System for Sustainability and Resilience, a tool designed to integrate multiple disciplines and foster collaboration among stakeholders. A key component of this approach was the creation of Facilitators—individuals tasked with bridging the gap between the scientific community and society. These Facilitators were trained through co-designed e-learning workshops, which also involved the development of basin- and barangay-scale hydrological models for real-time flood forecasting and climate change impact assessment.

The study emphasized the importance of tailored risk communication plans and workshop



designs. Facilitators developed twenty-one risk communication plans and workshop designs for fourteen barangays, considering various geographic, demographic, economic, and social features. These plans were intended to aid in public dissemination related to climate change adaptation and enhance community preparedness.

This approach demonstrates the efficacy of combining science-based knowledge with human resource development to strengthen flood resilience. By bridging the gap between scientific research and community engagement, such methodologies provide a practical framework for improving disaster literacy and fostering a more resilient society.

GIS-Based Risk Assessment of Structure Attributes in Flood Zones of Odiongan, Romblon, Philippines

Gacu, J. (2023). Flood triggered by heavy rains and typhoons leads to extensive damage to land and structures putting rural communities in crucial condition. Most of the studies on risk assessment focus on environmental factors and building attributes have not been given attention. The five most expensive typhoon events in the Philippines were recorded in 2008–2013, causing USD 138 million in damage costs. This indicates the lack of tools that would aid in the creation of appropriate mitigation measure/s and/or program/s in the country to reduce damage caused by typhoons and flooding. Hence, this study highlights a structure vulnerability assessment approach employing the combination of analytical hierarchy process, physical structure attributes, and existing flood hazard maps by the local government unit. The available flood hazard maps were layered into base maps, and building attributes were digitized using a geographic information system. The result is an essential local scale risk map indicating the building risk index correlated to the structural information of each exposed structure. It was



recorded that of 3094 structures in the community, 370 or 10.25% were found to be at moderate risk, 3094 (76.79%) were found to be high risk, and 503 (12.94%) were very high risk. The local government unit can utilize the resulting maps and information to determine flood risk priority areas to plan flood mitigation management strategies and educate people to improve the structural integrity of their houses. A risk map gives people an idea of what to improve in their houses to reduce their vulnerability to natural disasters. Moreover, the result of the study provides direction for future studies in the country to reduce loss and enhance structure resiliency against flooding. Gacu (2023) focuses on GIS-based risk assessment of structure attributes in flood zones, highlighting the importance of identifying vulnerable structures and prioritizing mitigation efforts. CityInsight can integrate similar risk assessment methodologies to identify areas prone to safety risks, such as flood-prone zones or high-crime areas. By providing users with localized safety ratings and risk assessments, CityInsight can empower individuals to make informed decisions and take proactive measures to enhance their safety and security in urban environments.

A Systematic Approach to Social Flood Vulnerability in Metro Manila

According to Dulawan, J. M. T., Imamura, Y., Konishi, T., Amaguchi, H., & Ohara, M. (2024). Assessing social vulnerability to floods (SVF) is a crucial component of integrated flood risk management, especially in flood-prone regions such as Metro Manila in the Philippines. This study presents a holistic framework to assess SVF, based on susceptibility, exposure, and resilience factors while recognizing its multi-dimensional, multi-attribute, and context-dependent nature. The assessment was applied to three cities of Metro Manila: Marikina, Parañaque, and Manila. Indicators were systematically identified using deductive and hierarchical approaches, and factor indices were derived using a combined Simple Additive



Weighting and Analytic Hierarchy Process, while the SVF index was estimated using the vulnerability relationship of the factors. The results exhibit a strong positive correlation with historical social flood damages and robustness across weight and normalization variations, with Marikina being the most vulnerable, followed by Parañaque and Manila. The city-level vulnerability profiles highlight salient determinants such as high population exposure in Parañaque, a relatively high population of susceptible social groups in Marikina, and significant disparity in flood control projects among the cities. Scenario analyses underscore the efficacy of flood countermeasures that reduce exposure and increase resilience. This study offers valuable insights and practical guidance for developing effective disaster reduction plans and policies in the study areas and similar regions by identifying disparities in vulnerability and understanding local conditions and drivers.

2021 Philippines Disaster Management Reference Handbook

National Disaster Risk Reduction and Management Council (NDRRMC) (2021), The Philippines, located in the Pacific Ring of Fire and typhoon belt, is prone to natural disasters like typhoons, earthquakes, floods, and volcanic eruptions. Typhoon Haiyan (2013) prompted significant improvements in disaster management, enhancing communication and coordination among national and international responders. These efforts continue to support recovery, especially post-COVID-19, amid growing climate change impacts such as rising sea levels and extreme weather events.

The Philippines has integrated disaster risk management and climate change adaptation into national frameworks, such as the National Disaster Risk Reduction and Management Act of 2010. The government is also investing in resilient and green infrastructure. In major disasters, international aid is coordinated through established systems, including those

strengthened by agreements like the 2014 Enhanced Defense Cooperation Agreement (EDCA).

System Development using Scrum Methodology

According to Hema, V. et al. (2020), Agile is an incremental and timeframe iterative approach. It supplies the software researchers with a working framework for traditional software development practices like waterfall model. Although the traditional models are best suited for small products where there are no changing requirements but not suitable for the products which have rapidly changing requirements and thus here, we recommend the SCRUM methodology. In this approach we can develop the software product by getting regular feedback from the customer through review meetings. So, whenever the customer requests the changes, we can upgrade the product with respect to those changes and can also develop the product more efficiently and rapidly.



Figure 1. Scrum Phases

The Evaluation of WebCost Using Software Usability Measurement Inventory (SUMI)

Zulkefli Mansor, Zarinah Mohd Kasirun, Saadiah Yahya, Noor Habibah Arshad (2021), The study highlights the importance of measuring user satisfaction with software interfaces and functionality, particularly before handing over a product to customers. The Software Usability Measurement Inventory (SUMI) was used to evaluate "WebCost," a standalone cost estimation



application developed using Java. By employing a set of SUMI-based questions, the evaluation revealed that WebCost offers user-friendly interfaces and produces accurate cost estimations.

The research emphasizes that user acceptance of a product depends significantly on its interface design and ability to deliver expected outcomes. Ignoring these aspects could result in low-quality software with minimal value to users. Therefore, tools like SUMI are essential for assessing software usability effectively, alongside alternatives such as User Acceptance Testing (UAT). This evaluation demonstrates how SUMI can measure interface effectiveness and ensure high-quality software development.

FEATURES OF THE SYSTEM

Geolocation Technology

Ambrosio-Santiago, B. et al. (2023) stated that nowadays, in Peru, there are several problems, where crime is one of the most prominent; we see this reflected daily in the various criminal acts that occur in our capital city, which generates fear, distrust and insecurity in the population. For this reason, we propose the development of a mobile application that can provide relevant information on crime incidents occurring in the district of Los Olivos. The application has the function of notifying users and district authorities about incidents of citizen insecurity to generate various lines of communication, which, by being more efficient, provide an immediate response to these criminal acts, thus reducing crime rates in the district of Los Olivos. In addition, this application promotes citizen participation and empowers the population to be part of the decisions in public policies, it will also allow the Police of the Jurisdiction to take the necessary actions since they will have updated and accurate information on their mobile device. For the development of the mobile application was chosen the Scrum methodology because it allows greater efficiency in the development stages. Finally, its structure will be made with the



Kotlin language, which will have the necessary Application Programming Interfaces (API's) and like Google Maps, to visualize and make use of these adding risk notifications about a crime in key locations, storing the information in a relational database for distribution through the virtual store of PlayStore applications.

Crime mapping report mobile application using GIS

Maghanoy, J. A. (2017). Criminal rates around Metro Manila have been increasing over the past years and it remains a challenge for Philippine National Police (PNP) to get accurate data because of some unreported crimes. As cited by PNP NCPRO officials that unreported crimes rate as much as 60% which presents a worsening peace and order situation in the country. As the new era strengthens the advances of technology, reporting of crime incidents in real time is now possible using smartphones. Different tools and techniques are being used to address the issues of crime but implementing the Geographic Information System (GIS) through mobile applications is not yet common. Thus, this paper intends to create a GIS-based android mobile application to report crime on location-based information. This will let the students be aware that a crime has taken place by using smartphones. This research will benefit both the local police and citizens living in Metro Manila. This project also aims to protect students in Ubel campus who are the usual target victims of the crimes. Maghanoy (2017) proposes a crime mapping mobile application using GIS technology to address increasing criminal rates in Metro Manila. This aligns with CityInsight 's objective of providing real-time safety information to users, as both initiatives leverage technology to enhance crime monitoring and reporting. By integrating crime data and incident reporting functionalities, CityInsight can serve as a valuable tool for both law enforcement agencies and citizens in monitoring and addressing safety concerns in urban areas.



Crowd Sourcing

Estellés-Arolas, E. (2022), Neighbours sharing information about robberies in their district through social networking platforms, citizens and volunteers posting about the irregularities of political elections on the Internet, and internauts trying to identify a suspect of a crime: in all these situations, people who share different degrees of relationship collaborate through the Internet and other technologies to try to help with or solve an offence. The crowd, which is sometimes seen as a threat, in these cases becomes an invaluable resource that can complement law enforcement through collective intelligence.

Enhancing Barangay Justice System Through the Development of a Web-Based Crime Monitoring Module

Antonio, L. (2020) states that a secure and safe environment is crucial for promoting investment and economic growth within a community. Peace and order are vital issues, as individuals must feel consistently safe to thrive. Public safety officials face the critical responsibilities of protecting people, property, and the environment, as well as resolving disputes. To enhance their effectiveness, recent studies have explored technological solutions aimed at improving public safety management.

One notable study developed a Web-based Crime Monitoring Module designed to strengthen the barangay justice system through improved record-keeping and management of community offenses. This system introduces a more efficient method for monitoring offenses and blotter cases in selected barangays, aiming to enhance the administration of peace and order. A key feature of the system is its crime-mapping capability, which integrates Google Maps to allow administrators to tag reported crimes and blotter cases geographically. This functionality



facilitates the identification of crime hotspots within the community.

Synthesis

The study by Mojica et al. (2017) visualized the spatial patterns of murder and physical injury in Metro Manila, Philippines, using conditional choropleth maps. The researchers investigated the relationship between these crime rates and various demographic variables while accounting for potential spatial autocorrelation through spatial lag models. Their findings revealed that both crime rates tend to cluster in the northern cities of Metro Manila, with significant spatial lag coefficients identified only for physical injury rates, indicating a positive city-level spatial dependence. Additionally, demographic factors such as population density, percentage of young males, education, marriage, and immigration were found to be associated with both crime rates. These results suggest that crime monitoring systems should incorporate these demographic indicators to enhance resource allocation and program planning for more effective crime prevention and security management.

The studies by Mojica et al. (2017) and Estellés-Arolas (2022) explore different yet complementary approaches to enhancing crime prevention and law enforcement. Mojica et al. (2017) focus on the spatial and demographic analysis of crime patterns in Metro Manila, using conditional choropleth maps and spatial lag models to identify clusters of murder and physical injury rates in the northern cities of the region. They highlight the significance of demographic factors such as population density, the percentage of young males, education levels, marriage rates, and immigration in influencing these crime rates. This data-driven approach suggests that Integrating demographic indicators into crime monitoring systems could improve resource allocation and crime prevention strategies.



In contrast, Estellés-Arolas (2022) examines the role of collective intelligence and crowdsourcing in crime prevention, emphasizing how individuals collaborate through digital platforms to share information about local crimes, election irregularities, and criminal suspects. The study argues that, while the crowd is sometimes perceived as a potential threat, it can be an invaluable resource that complements traditional law enforcement efforts. By illustrating how online collaboration can assist in solving offenses, this study underscores the potential of collective efforts in crime prevention.

When combined, these studies offer a holistic view of crime prevention. Mojica et al. provide insights into the spatial and demographic factors influencing crime, which can guide targeted interventions, while Estellés-Arolas highlights the power of crowd-based collaboration in supporting law enforcement. Together, they suggest that a blend of data-driven analysis and collective intelligence could lead to more effective and comprehensive crime prevention and security management strategies.

The studies on CityInsight, "iReport Mo Kay Mayor," and the PNP Crime Incident Reporting System each address different aspects of public safety and governance in the Philippines but share a common goal of enhancing efficiency, transparency, and responsiveness through digital solutions.

CityInsight focuses on improving safety and emergency response within Urdaneta City by providing a web-mobile application that allows residents to tag hazardous locations and report emergencies. This platform aims to boost user confidence in exploring different areas by offering features such as hazard location tagging, emergency alerts, and real-time notifications about unsafe areas. By enabling direct reporting of incidents and alerts to administrators, CityInsight



facilitates prompt action and coordination with government units like the Philippine National Police (PNP) and the City Disaster Risk Reduction Management Office (CDRRMO). The study is specifically focused on evaluating the development and functionality of this application within Urdaneta City.

"iReport Mo Kay Mayor" enhances local governance and public service delivery by offering a digital platform for citizens to report various local issues—such as crime incidents and infrastructure problems—directly to their local government offices. This system streamlines the reporting process, reduces delays, and improves transparency by providing a direct communication channel between citizens and their mayors. It supports data-driven decision-making by enabling local governments to track and manage reports effectively, fostering greater public trust and engagement in governance. The platform's scope is centered on improving local government responsiveness and accountability.

The PNP Crime Incident Reporting System modernizes law enforcement by automating crime reporting processes, increasing efficiency, accuracy, and reliability. This system addresses the limitations of manual documentation by allowing police officers to log and manage incidents more swiftly and accurately. It enhances data quality, supports data-driven policing, and improves resource allocation. By enabling online reporting of crimes and suspicious activities, the system fosters community involvement in public safety. Additionally, it ensures real-time communication and coordination among police units while maintaining the security and confidentiality of sensitive crime data.

All three studies highlight the transformative potential of digital solutions in public safety and governance. CityInsight and "iReport Mo Kay Mayor" both aim to enhance public



engagement and responsiveness, though in different contexts—CityInsight focuses on safety and emergency response within a specific city, while "iReport Mo Kay Mayor" targets broader local governance issues. The PNP Crime Incident Reporting System, on the other hand, specializes in modernizing crime reporting and data management within law enforcement. Each platform contributes to creating a more modern, efficient, and transparent system for managing public safety and community concerns, reflecting a broader trend towards digitalization in governance and public service. Together, these studies demonstrate how digital innovations can address various challenges in public administration, ultimately leading to more responsive, transparent, and effective governance frameworks.

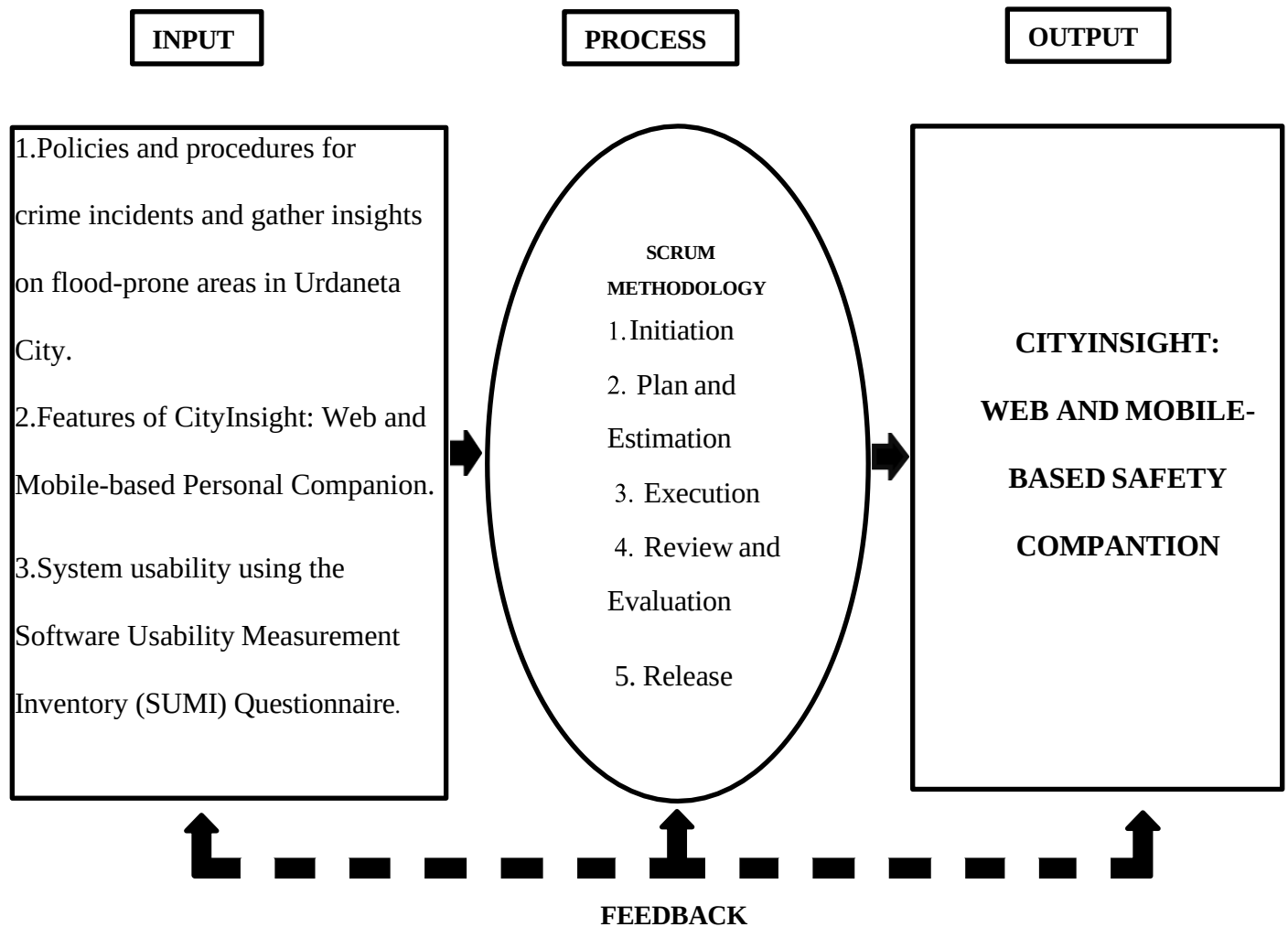
Project Framework

The CityInsight project will utilize a hybrid project management approach, combining the structure of Scrum with the flexibility of Agile methodologies. This ensures a clear roadmap with defined phases (initiation, planning, development, launch, and maintenance) while allowing for iterative development and user feedback integration during the pilot launch. The project life cycle will be divided into distinct phases with specific deliverables and milestones. Throughout the development process, project management software and communication tools will keep the team organized and stakeholders informed. A version control system will facilitate collaboration among researchers, and user feedback mechanisms will be crucial for app improvement. Risk management will be proactive, identifying potential issues and developing mitigation strategies. Stakeholder management will involve clear communication plans and regular updates to ensure everyone's interests are considered. Monitoring and evaluation will be continuous, tracking progress, user engagement, and the app's impact on safety



perception.

The diagram outlines the structure of the research project, which involves input consists of a (1) Develop and Define the user safety needs, (2) Development of CityInsight: Web and Mobile-based Personal Safety Companion and (3) Evaluate Impact and Usability. These tasks will be executed using various stages of the Scrum Methodology. The anticipated outcome of this project is a system for "CityInsight: Web and Mobile-based Personal Safety Companion" tailored to the people of Urdaneta City, Pangasinan.



Feedback Figure 1: Input-Process-Output (IPO) Model for Research Project



CHAPTER III

METHODOLOGY

This Chapter provides a view of methods that will be applied in this project. Also, project design, population and locale of the study, data instrumentation, data gathering procedures, and data analysis are presented in this chapter.

Project Design

The research study will employ developmental and descriptive research methodologies to design, develop, test, and evaluate CityInsight, a web-mobile application designed to provide insights into crime rates and safety through crowdsourced information. The integration of these methodologies will be crucial in achieving significant findings during the study.

Descriptive research will be utilized for Objective 1, Objective 2, and Objective 4 of the study, incorporating both qualitative and quantitative methodologies to provide a comprehensive description, explanation, and analysis of CityInsight's functionality, usability, and its impact on crime rates and safety. The research process began with a planning phase, where the objectives were clearly defined, and the appropriate research tools were developed. This included the creation of a structured survey questionnaire instrument, which was meticulously designed to gather detailed information from respondents. To gain deeper insights, the study also incorporated interviews with key stakeholders and residents to capture qualitative data on safety concerns and perceptions within specific barangays in Urdaneta City. Additionally, the Likert Scale methodology was employed to quantitatively assess the level of safety concerns, providing a robust analysis of the issues at hand.



Initiation - In the initiation phase, the Scrum team, consisting of a Scrum Master, Product Owner, and Development Team, is established at the project's initiate phase. To guarantee that everyone in the team agrees with the project's objectives, the team starts by creating a distinct project vision. To obtain specific needs and information about the goals and limitations of the project, this involves carrying out stakeholder interviews. A complete vision can be formed and aligned with stakeholder expectations with the aid of these interviews. To help with the process of creating a product backlog, a prioritized list of features and tasks needed to realize the vision, meetings are planned. In order to ensure that every detail is covered and given the proper priority, the team breaks down the project requirements into manageable tasks during these meetings. In order to establish realistic deadlines and milestones, release planning meetings are held to lay out the timeframe for delivering the various project components. To identify possible difficulties, a preliminary risk assessment is carried out. Meetings are then arranged to talk about and create measures for mitigating the risks. These sessions also result in the establishment of the Definition of Done (DoD), which establishes precise quality requirements for finished work. Effective information exchange is facilitated by the establishment of communication channels, which guarantees that the team stays informed and involved throughout the project. This stage establishes the project's course and makes sure that everyone in the team is ready for the development phase.

Plan and Estimation - During the implementation phase, the researchers of the application will use Flutter to construct the system's mobile part, with the objective to deliver smooth and user-friendly experiences. In terms of online applications, they will plan to use PHP to build strong and dynamic functionality on the web platform. This combination is expected to result



in a flexible and effective system that responds to a variety of users across devices and platforms.

Execution - The team employs a methodical approach to development, executing the project in stages and providing frequent updates and reviews to monitor advancement and tackle any challenges that arise. The Flutter framework is used to create a high-performing mobile application for the CityInsight project. Flutter's features provide a fluid user experience from a single codebase for Android platform, resulting in a dynamic and aesthetically pleasing user interface. PHP is used to provide a comprehensive and adaptable web interface that improves the application's functionality and accessibility on many devices. Reliable server-side processing and seamless integration with the mobile front end are ensured by the usage of PHP. Daily team conferences and changes to the project backlog are used to monitor the development process, guaranteeing timely delivery and efficient administration of tasks.

Review and Evaluation - During the review and retrospective phase, researchers will conduct usability testing to evaluate the performance and functionality of the program. This assessment will determine its effectiveness and verify its alignment with end-users' expectations. Through a thorough evaluation and reflection, the team will aim to identify areas for improvement and ensure that the program meets the desired standards of usability and functionality.

Release – During the release phase, the researchers will release the initial release of the program, which will be an important turning point in the development process. This step comprises launching the system and making it available to the target user population. Before this, there will be a time of planning, involving extensive configuration, testing, and validation



to make sure that the application is ready to go operational. With all this security in place, the release phase provides the starting point for users to interact with the program, resulting in actual events usage and impact. This phase not only marks the end of development, but also allows for user involvement and data collection. As users explore the application's functionality, their input is essential for future improvements and changes.

Population and Locale of Study

A. Population

The study will be conducted around Urdaneta City, Pangasinan. To ensure a comprehensive understanding of the functional requirements of the proposed system, we will use purposive sampling to select 30 participants from Urdaneta City. This method allows us to intentionally include key stakeholder groups—residents, students, police officers, and NDRRMC officers—who possess specific knowledge and perspectives relevant to the system. By focusing on these targeted groups, we ensure that all critical viewpoints are considered, providing a nuanced and informed basis for the system's design. The selected sample size of 30 participants allows for in-depth qualitative analysis. By focusing on a smaller, targeted group, we can gather rich, detailed insights that are crucial for understanding the specific functional requirements of the proposed system.



Locale of the Study

The researchers aim to implement the “CityInsight: Web and Mobile-based Safety Companion” system in Urdaneta City, Pangasinan, with the goal giving customers real-time safety insights will allow them to navigate unfamiliar surroundings with more understanding and confidence.

Objective #	Police In Charge	NDRRMC Respondents	Urdaneta Residents	Urdaneta City Students	Total
1	1	1	0	0	2
2	5	5	10	10	30
4	5	5	10	10	30

Table 1. *Distribution of Respondents*

For Objective 1, The police in charge of Urdaneta City PNP and an NDRRMC personnel will be interviewed to obtain the necessary details and fulfill the objective.

For Objective 2, To obtain the appropriate user requirements, the respondents will consist of residents, students who travel from their home to school, PNP Officials, and CDRRMC Personnel in Urdaneta City with the total of 30 participants.

For Objective 4, the researchers will select thirty (30) respondents for the usability testing which are potential end users from Urdaneta City that consists of PNP Officials, CDRRMC Personnel, Residents, and Students.



Data Instrumentation

The following instruments were utilized by the researchers to gather additional information that the respondents require.

For Objective no. 1, Using structured interview as a data collection method, the researchers focus on the policies and procedures for accurately disseminating crime incident data. It will also aim to gather detailed information about flood-prone areas in Urdaneta City. The researchers will ensure that all collected information is precise and reliable as this structured approach will provide a solid factual foundation for developing the system, ensuring that it is based on comprehensive and accurate data. Refer to Appendix A and B for the structured interview guide.

For Objective no. 2, Based on ratings obtained from a survey checklist, this objective involves identifying and prioritizing the system's features. It focuses on defining the functional requirements of the "CityInsight: Web and Mobile-based Personal Safety Companion" for crime and safety. By systematically evaluating user feedback through this checklist, the researchers will ensure that the development process aligns with the users' unique functional needs, resulting in a system that effectively addresses their requirements and enhances personal safety. Refer to Appendix C for the survey checklist guide.

For Objective no. 4, The researchers will use a survey checklist to gather and analyze data provided by the CityInsight system to get insight into patterns of usage and operational efficiency. The obtained data will be assessed against five essential parameters: (a) Efficiency, (b) Convenience, (c) Usefulness, (d) Environmental impact, and (e) Learnability. This will help

with the continual growth of urban planning and policymaking, ultimately improving the system's overall efficacy and sustainability. Refer to Appendix D for the survey checklist guide.

Data Analysis

In the development of the system. The researchers used the following tools for data analysis.

For Objective 1, The interview findings will be analyzed using thematic analysis and presented through a flowchart diagram to evaluate the policies and procedures currently in place for responding to various crime incidents and managing flood-prone areas in Urdaneta City. This includes identifying key themes, procedural gaps, and insights through qualitative data analysis.

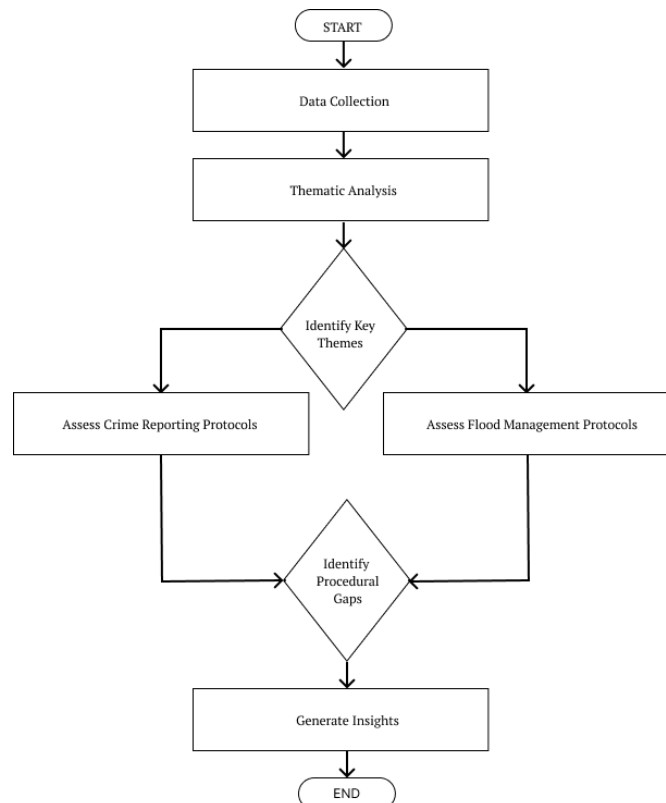


Figure 2. Flowchart Diagram for Objective 1



The process begins with collecting interview data and then transcribing the recordings into text. Next, responses are coded by labeling and organizing the text to identify key themes and patterns related to the policies and procedures currently in place for responding to various crime incidents and managing flood-prone areas in Urdaneta City. Procedural gaps are analyzed by assessing the effectiveness of current protocols. Insights are extracted from this analysis to provide actionable conclusions and recommendations, concluding the process with findings prepared for presentation.

For Objective 2, The researchers will analyze the functional requirements by counting how frequently each feature is requested. Features requested by 50% or more of the 30 respondents (which equates to at least 15 people) will be prioritized for development. Features requested by fewer than 50% of respondents may be considered optional. This ensures that development efforts focus on the features that users deem most important

For Objective 4, the results from the survey on usability test will be tallied using frequency count and mean, and interpreted using the Likert scale below

Mean Score Range	Descriptive Equivalent	Descriptive Interpretation
4.30 – 5.00	Strongly Agree	Usable
3.50 – 4.29	Agree	Usable
2.50 – 3.49	Moderately Agree	Usable
1.70 – 2.49	Disagree	Not Usable
1.00 – 1.69	Strongly Disagree	Not Usable

Table 2. *Likert Scale for Usability Testing*

Responses falling between 1.00 and 2.49 signify that the system fails to meet user needs and is deemed unusable, while for positive questions, a score of 2.50 or higher



indicates the system meets user requirements and is usable, and for negative questions, it indicates fewer problems and better usability.

Proposed Implementation Plan

"CityInsight: Web and Mobile-Based Personal Safety Companion". By providing customers with real-time safety insights, our application empowers them to navigate unfamiliar surroundings in Urdaneta City with greater understanding and confidence.

Built on Flutter for seamless cross-platform performance, with PHP handling backend functionalities, and firebase for strong database storage, CityInsight ensures a smooth and reliable user experience.

We leverage Figma for UI/UX design, allowing us to create high-fidelity prototypes and design simulation. Figma's user-friendly interface, collaborative features, and cloud-based storage facilitate efficient iteration on designs, incorporating feedback from team members and stakeholders seamlessly.

Once the initial prototype is completed, rigorous testing will be conducted in Urdaneta City, ensuring that CityInsight meets the needs and expectations of our users in real-world scenarios.

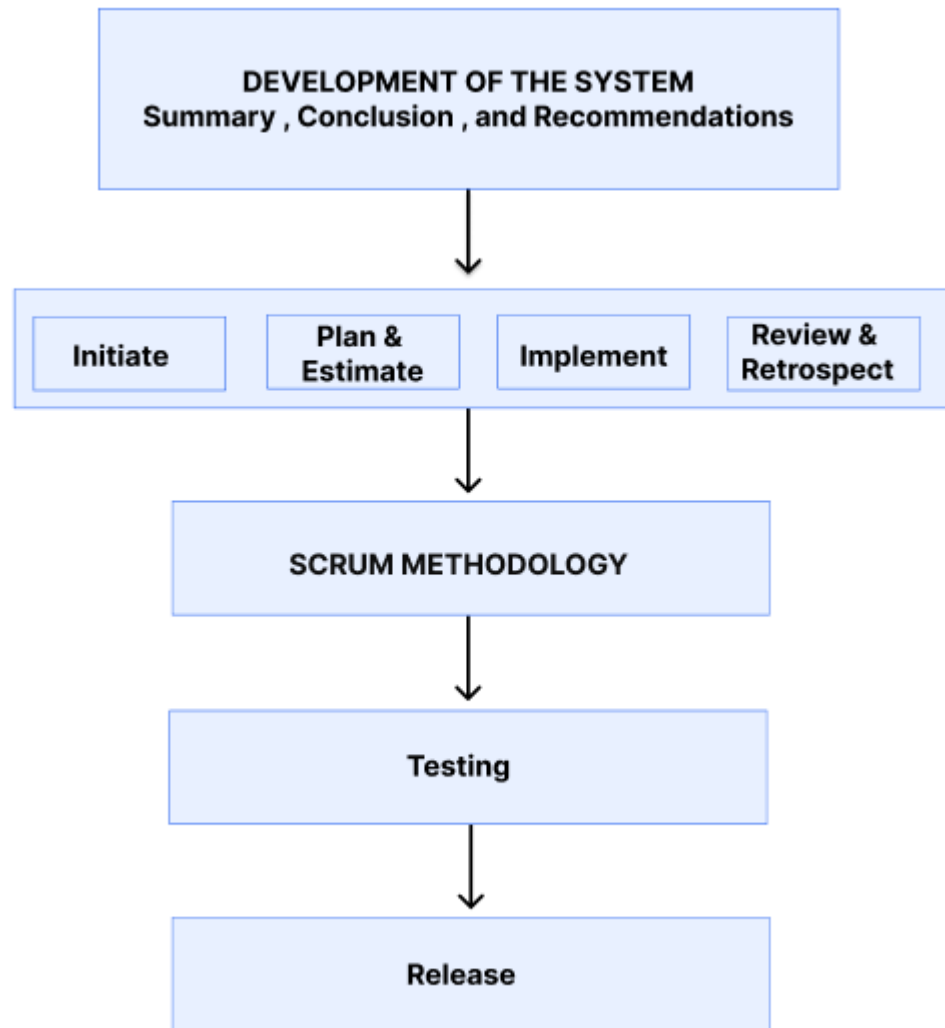


Figure 3. Flowchart Diagram for Proposed Implementation Plan



CHAPTER 4

RESULTS AND DISCUSSIONS

This chapter includes the analysis and interpretation of the findings of the study. It also presents the existing process and developed system using data modeling such as flowchart and diagrams.

Processes of the CityInsight Application

Using Scrum Methodology, the proponents conducted a structured interview with the PNP, CDRRMC, and Students. Documentary analysis of forms and reports was also done along with observation within the city. As a result, the proponents determined the different processes underwent by the Police Stations using the system which is represented by the Flow Charts in the following figures.

Traditional Process in Terms of Responding to Natural Disasters

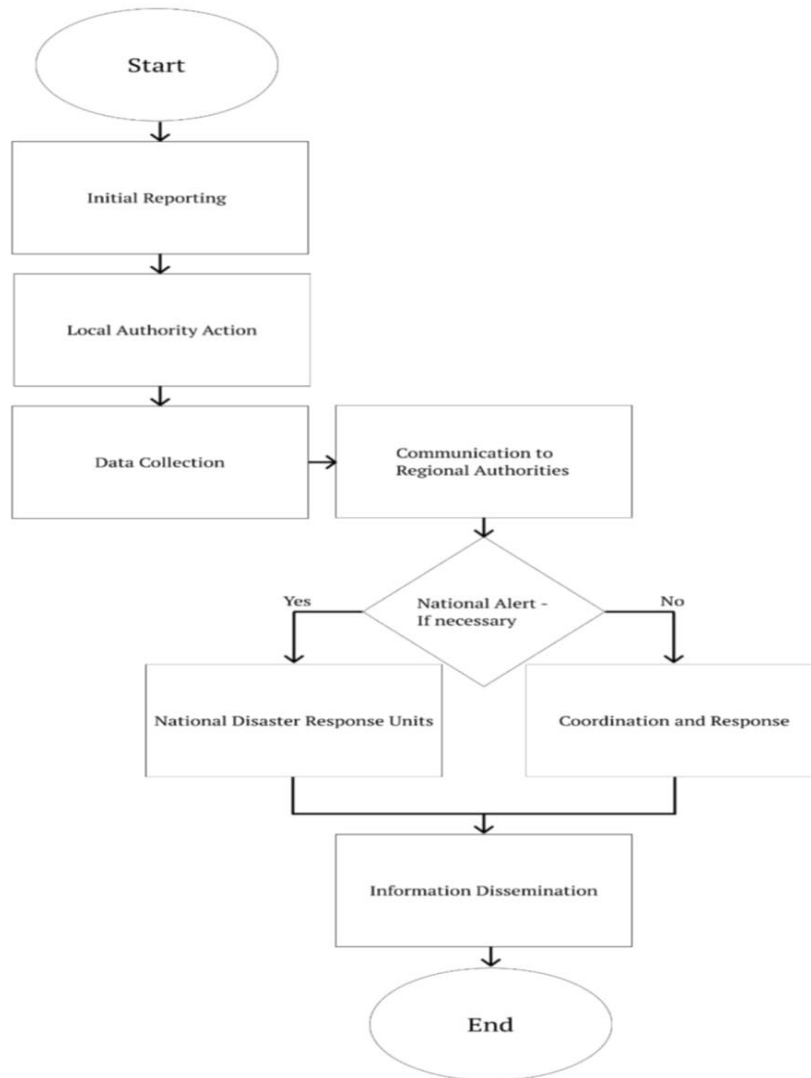


Figure 4. Flow Chart for the Traditional Process in Terms of Responding to Natural Disasters

Figure 4 shows the traditional disaster response process of the City Disaster Risk Reduction and Management Council (CDRRMC). It begins with the issuance of disaster

alerts to inform the community. After the alert, an assessment of the disaster's impact is conducted to evaluate damage and risks. Based on the assessment, response mechanisms are activated, including mobilizing resources and deploying response teams. This is followed by rescue and relief operations to ensure the immediate safety and well-being of affected individuals. Finally, the process transitions to recovery and rehabilitation, focusing on rebuilding and restoring normalcy. The flowchart highlights the linear progression from alert to recovery.

Traditional Process in Terms of Generic Emergency Response

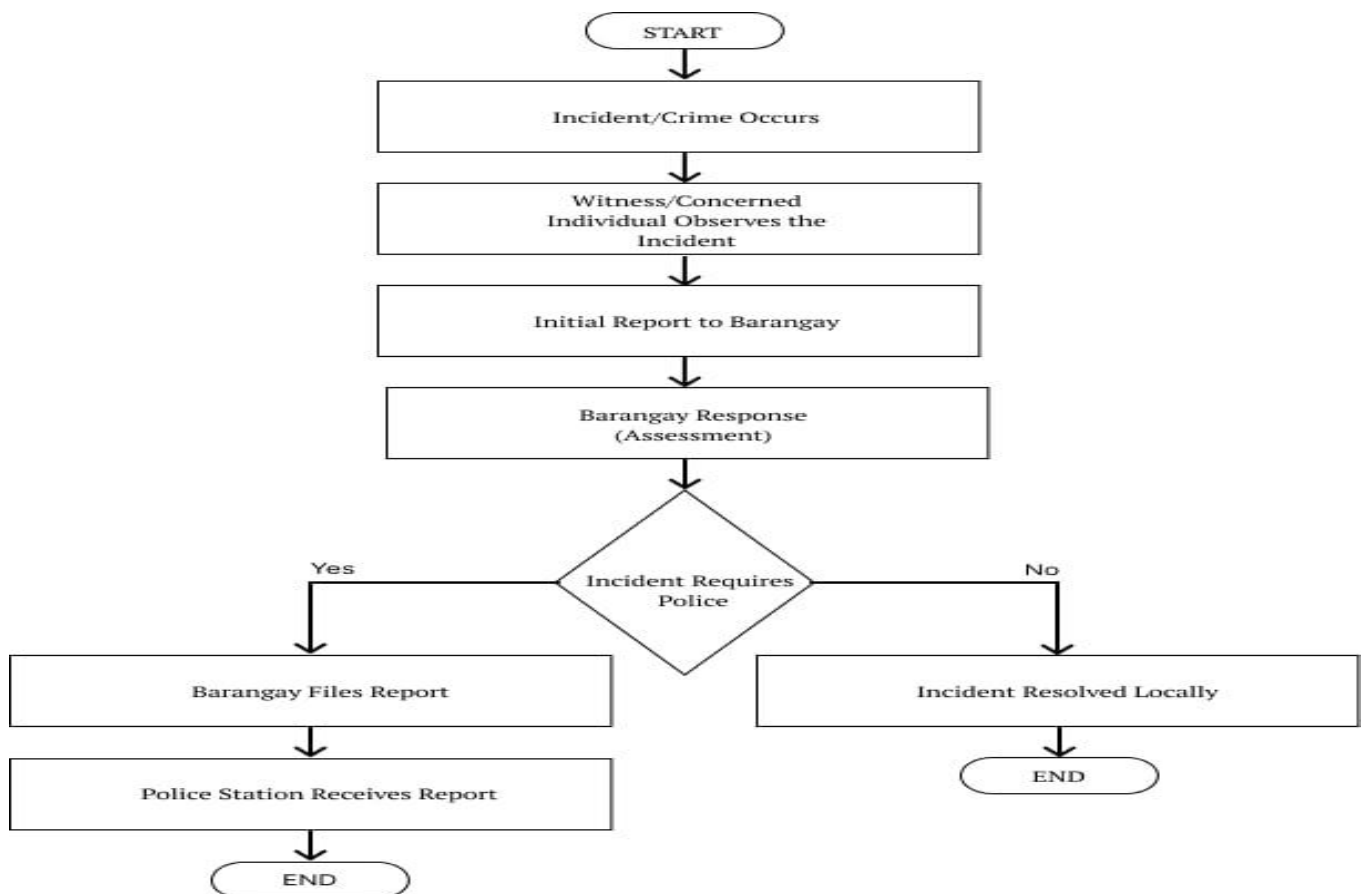


Figure 5. Flow Chart for the Traditional Process in Terms of Generic Emergency Response

Figure 5 shows a general emergency response process. It starts with the observation of an event, such as a natural disaster or major incident. The event is then reported to the appropriate authorities for action. The local authority assesses the situation and collects relevant data before communicating with regional authorities. A decision point determines whether a National Alert is necessary. If a National Alert is issued, National Disaster Response Units are activated; if not, the process shifts to coordination and response efforts. After critical actions are taken, information is disseminated to stakeholders, and the process concludes with coordination and response activities. This flowchart emphasizes decision-making and structured response actions.

Traditional Process in Terms of Incident/Crime Reporting

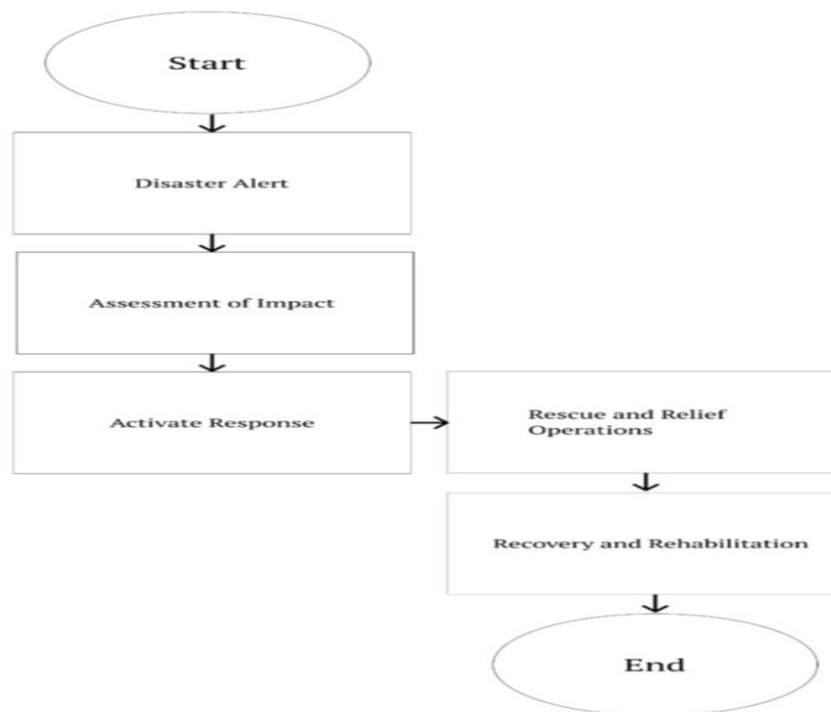


Figure 6. Flow Chart for the Traditional Process in Terms of Incident or Crime Reporting

Figure 6 shows that the Traditional Process of reporting an incident or crime in the Urdaneta City, Pangasinan. It begins when the incident occurs, followed by a witness

reporting it to the Barangay. The Barangay assesses the situation and decides whether police involvement is necessary. If so, the Barangay files a report with the local police station. If not, the incident is resolved locally within the Barangay. This flowchart highlights the decision points such as the need for police involvement.

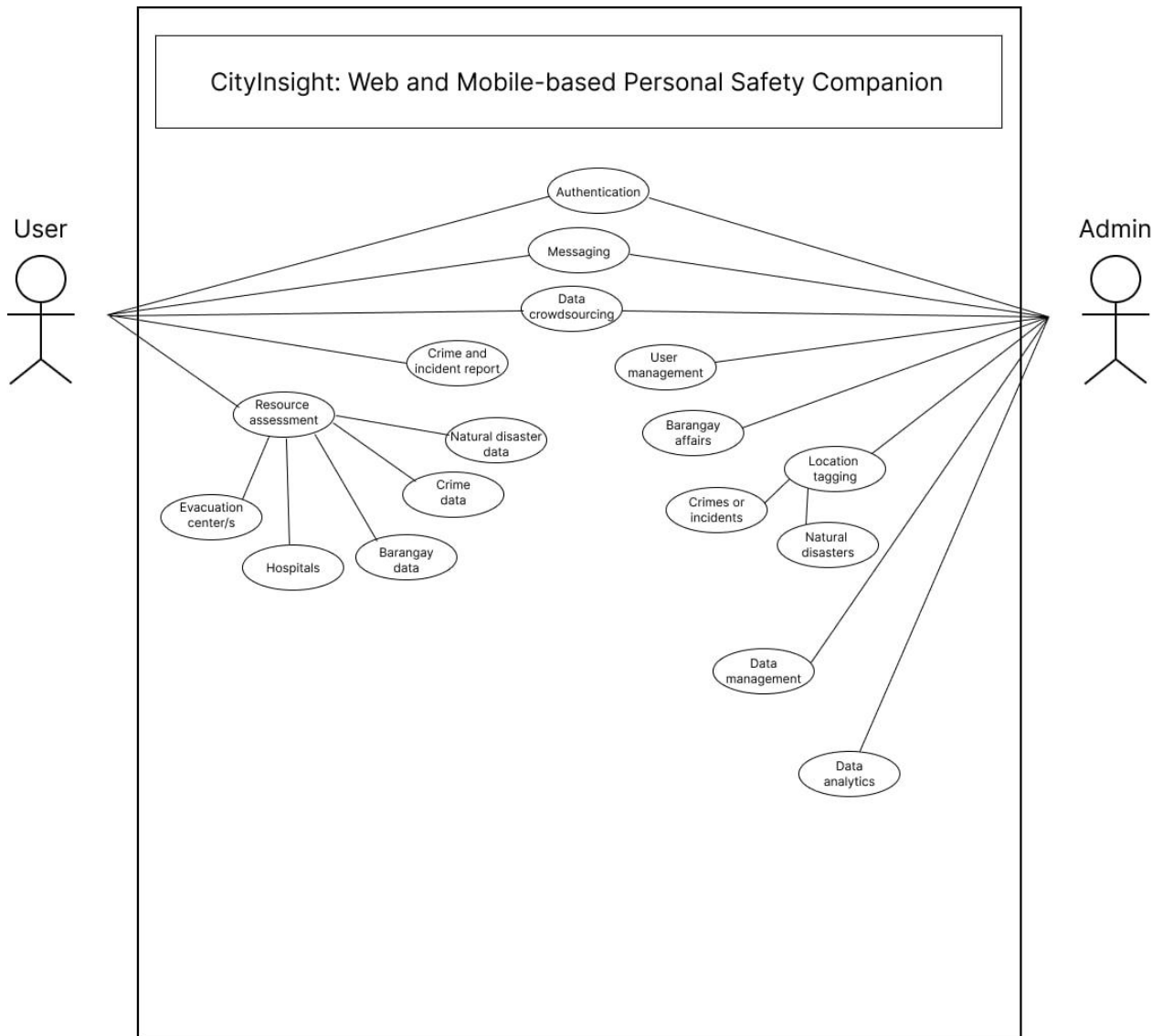


Figure 7. Use Case Diagram of the Application

Figure 7 demonstrates the use case diagram for the suggested system, illustrating the actions each user can perform actors in the proposed system include residents,

emergency responders, and the system administrator. For crime-related scenarios, residents can report incidents, tag locations, and request assistance, while responders provide support and manage reports. In flood-related scenarios, residents can report flooding, locate evacuation centers, and request rescue if needed, while responders coordinate evacuations and manage incoming reports.

Urdaneta City Flood Map

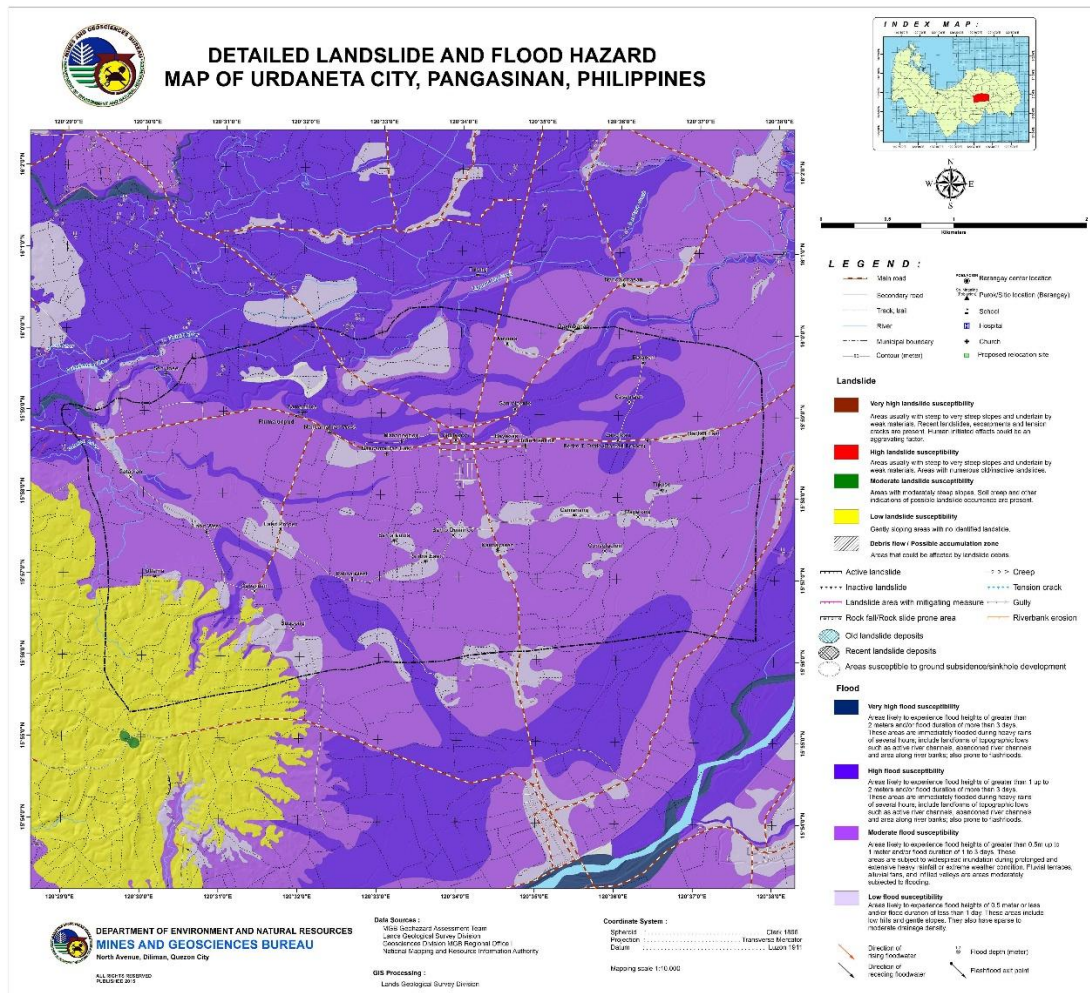


Figure 8. Urdaneta City Flood Map



Figure 8 illustrates the flood map for Urdaneta City, providing a visual representation of areas susceptible to flooding. The map uses color coding to distinguish various levels of flood risk across the city, making it easy to identify regions that are either at low, moderate, or high risk of flooding. The map outlines key streets, neighborhoods, and landmarks in Urdaneta City, allowing residents and emergency responders to see which areas may be affected by rising water. The data presented in the map is based on real-time flood reports, showing the extent and severity of flooding in different locations. This flood map is an essential tool for understanding and managing flood risks, enabling individuals and authorities to make informed decisions during flood events.

Functional Requirements Features

The system's capabilities and functional aspects that each user can use are displayed in the tables below. To aid the user in understanding the developed application, the table also provides a description of how each feature functions.

Table 3. Functional Requirements of CityInsight

Requirements	Description	User Type	Frequency	Percentage
Location Tagging	Location tagging lets users and admins mark specific points of interest or incidents on a map, enabling real-time identification of key locations to improve situational awareness and response coordination.	User and Admin	30	100%



Crime or Disaster Mapping	Crime or Disaster Mapping User and Admin	25	83.33%
Mapping	visualizes incidents and hazards on an interactive map, helping users and admins identify patterns and high-risk areas. This feature enhances situational awareness, allowing for better resource allocation and informed decision-making during emergencies.		
Crime Heat Maps	Interactive map feature that Admin visually highlights areas with higher concentrations of crime using color coding. The map will display crime data by location, with different colors representing various levels of crime intensity, such as low, medium, and high. This feature will make it easier for users to identify hotspots and understand crime patterns in specific areas.	21	70%.



Real-Time Location	Real-Time Location provides live tracking of users or incidents, enabling immediate location updates for accurate situational awareness.	User and Admin	20	66.67%
Incident Reports	Allows users to upload and view images or videos of crime scenes to raise awareness and assist investigations. Users can add details such as the type of crime, date, time, location, and a brief description with each media upload. These files will be stored securely and made accessible based on user roles and permissions.	User	20	66.67%
Analytics	Analytics provides weekly or monthly insights into crime, flood, and incident data at the barangay and town levels, keeping users informed with up-to-date information.	Admin	18	60%



One Click Emergency	One Click Emergency allows users to quickly report emergencies or request assistance with a single tap, streamlining the process for immediate response and support.	User and Admin	18	60%
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Table 3. *Functional Requirement Features*

Table 3 shows the functional requirements of the system designed for user and admin. This app will provide essential features like location tagging, crime or disaster mapping, crime heat maps, real-time location, incident reports, analytics and one-click emergency. The responses indicate the frequency and percentage of users who prioritized these functions, based on a total of 30 respondents.

Developed CityInsight: Web and Mobile-based Personal Safety Companion

The researchers were able to finish the project on time thanks to the researchers' adoption of the SCRUM approach, which guarantees to deliver clear and ordered descriptions and manageable projects.

Initiate. Plan and estimate, implement, review and retrospect, and release are the five SCRUM phases. The scrum methodology's iterations of the phases can aid developers in having a better, more manageable project. Initiate. The developers created the project vision and identified the stakeholders for the project.

Plan and Estimate. During the implementation phase, the researchers of the application will use Flutter to construct the system's mobile part, with the objective to deliver smooth and user-friendly experiences. In terms of online applications, they plan to



use PHP to build strong and dynamic functionality on the web platform. This combination is expected to result in a flexible and effective system that responds to a variety of users across devices and platforms.

Implement. The implementation phase of CityInsight involves the development of both mobile and web applications. Android Studio is used to build the cross-platform mobile app using Flutter, ensuring consistency across devices. Visual Studio Code is employed for the web application, enabling seamless integration with the mobile app. PHP handles server-side functionalities, while Firebase provides robust database storage and real-time capabilities. Figma is utilized for creating high-fidelity prototypes, facilitating efficient design iterations and collaboration. These tools and technologies collectively contribute to the development of a reliable and user-friendly solution that enhances safety and empowers individuals in Urdaneta City.

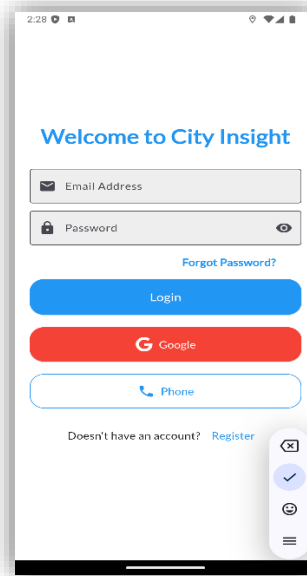
Review and Retrospect. During this phase, the team conducted a thorough review of the completed work, identifying and fixing any issues. This involved testing and debugging to ensure the application's optimal performance and reliability.

Release. The release of the application would be done within Urdaneta City to be handled by the Police and NDRRMC office's device. The end users can download the app to their personal smartphones or directly use the website on their personal computers. The app will be pre-installed on devices managed by the Urdaneta City Police Office and the NDRRMC based in Urdaneta City.



Screenshots of the Application

Plate # 1
Login UI of the Mobile Application for End Users



This interface shows the login page of the end users. The registered Email Address and Password of the end users can be used in this page to be able to login within the app.

Plate # 2
Home Screen of the Mobile Application for End Users

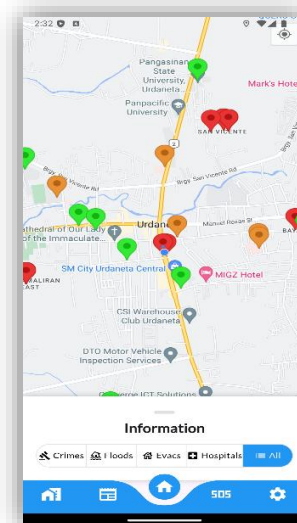


Plate 2 shows the Homescreen of the mobile application. The Homescreen includes the real-time location, navigate evacuation centers and hospital using polylines and the map that shows the information about crimes, floods, and evacuation center.

Plate # 3
Barangay List of the Mobile Application for End Users

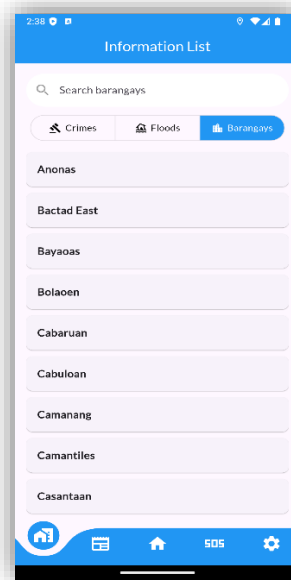


Plate 3 shows the Barangay List of the mobile application. The Barangay List includes the buttons that can redirect the end users to crimes, flood, and the list of Barangays.

Plate # 4
Crime Data of the Mobile Application for End Users



Plate 4 shows the Crime Data of the mobile application. The Crime Data includes the graph, type of crimes, details and count of crimes in the Barangay.

Plate # 5
Information Details of the Mobile Application for End Users

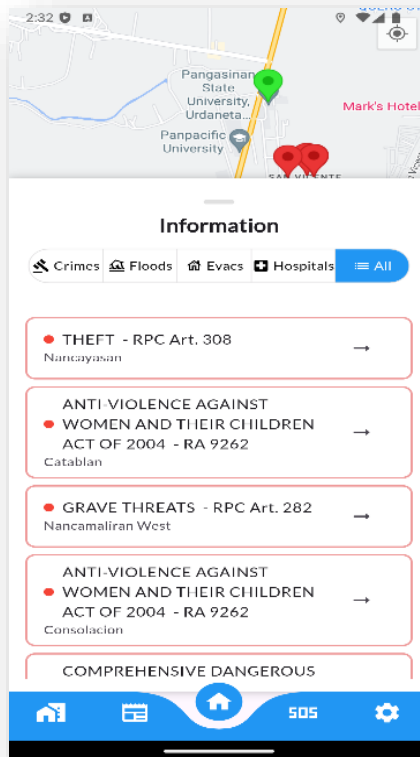


Plate 5 shows the Information Details of the mobile application. The Information Details include the details of crimes and the barangay that are affected by the crime.



Plate # 6
Report Incident of the Mobile Application for End Users

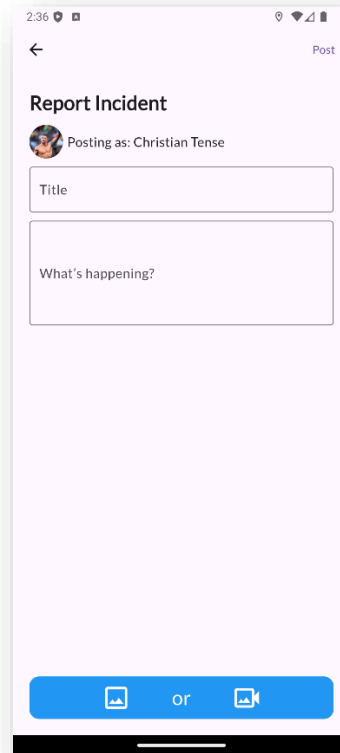


Plate 6 shows the Report Incident of the mobile application. The Report Incident include the report of crime or flood incident the end user can upload images and videos.

Plate # 7
Post Status of the Mobile Application for End Users

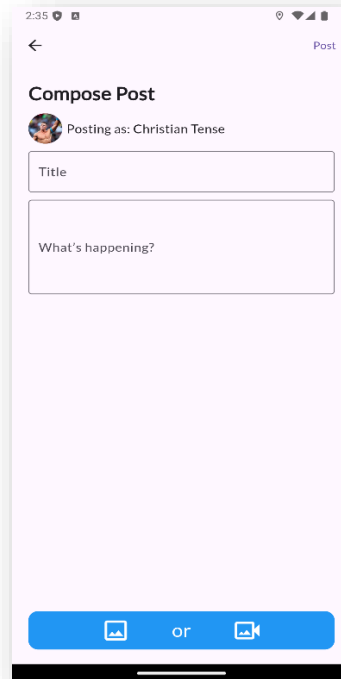


Plate 7 shows the Post Status of the mobile application. The Post Status include the posting of crime or flood incident the end user can upload images and videos.

Plate # 8
One Tap Alert of the Mobile Application for End Users

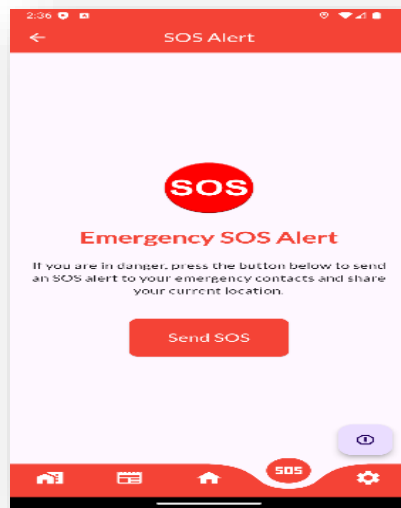


Plate 8 shows the One Tap Alert of the mobile application. The One Tap Alert include the button allows users to quickly report emergencies or request assistance with a single tap, streamlining the process for immediate response and support.

Plate # 9
Crowdsourcing of the Mobile Application for End Users

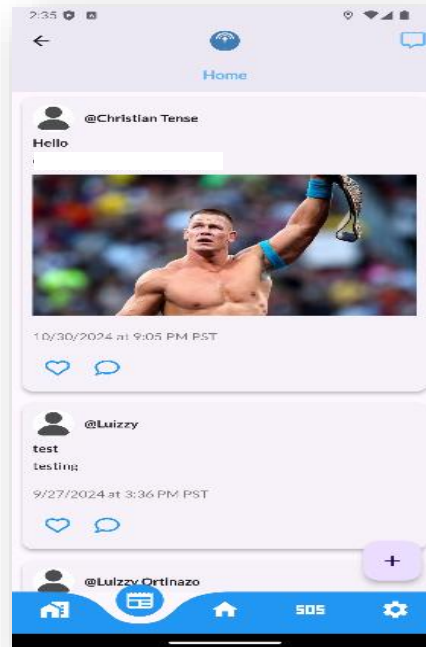


Plate 9 shows the Crowdsourcing of the mobile application. The Crowdsourcing include the posting of other users, and the other users can react and comment to the post.

Plate # 10
Chat Screen of the Mobile Application for End Users

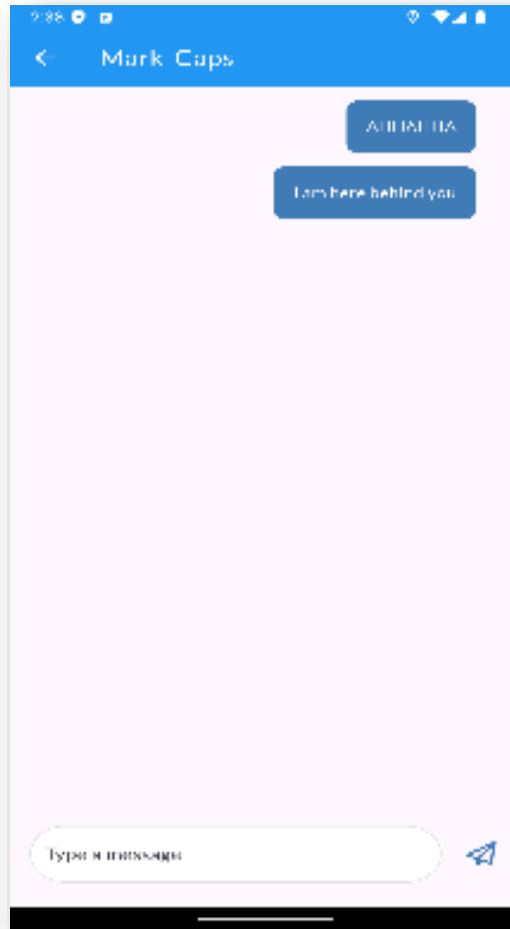


Plate 10 shows the Chat Screen of the mobile application. The Chat Screen include the chatting or messaging the other users.

Plate # 11
Message Screen of the Mobile Application for End Users

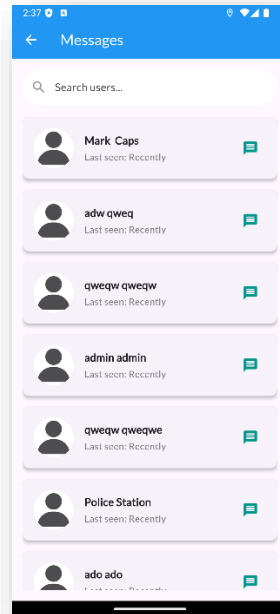


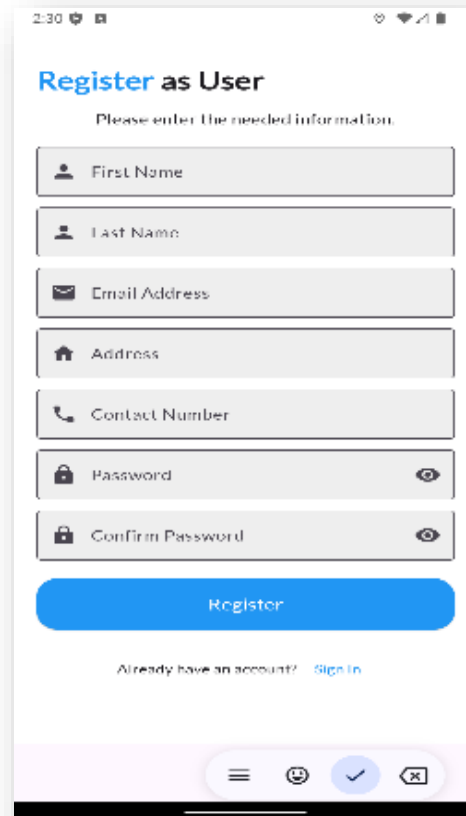
Plate 11 shows the Message Screen of the mobile application. The Message Screen includes the messaging the other users and the list of messages.

Plate # 12
Profile Screen of the Mobile Application for End Users



Plate 12 shows the Profile Screen of the mobile application. The Profile Screen include the Account details of the users and a button that can update of user's information.

Plate # 13
Register Screen of the Mobile Application for End Users



2:30

Register as User

Please enter the needed information.

First Name

Last Name

Email Address

Address

Contact Number

Password

Confirm Password

Register

Already have an account? [Sign In](#)

Plate 13 shows the Register Screen of the mobile application. The Register Screen includes the Information of the user to register.



Plate # 14
Forgot Password of the Mobile Application for End Users

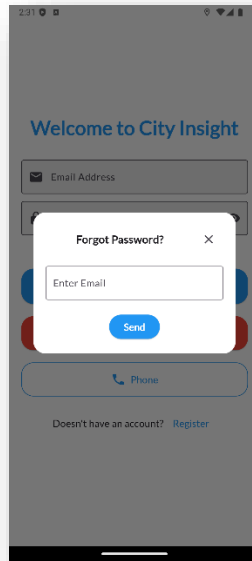


Plate 14 shows the Forgot Password of the mobile application. The Forgot Password includes the provided email, link will be sent to your email, and the user can change password.



Plate # 15
Login UI of the Web Application for End Users

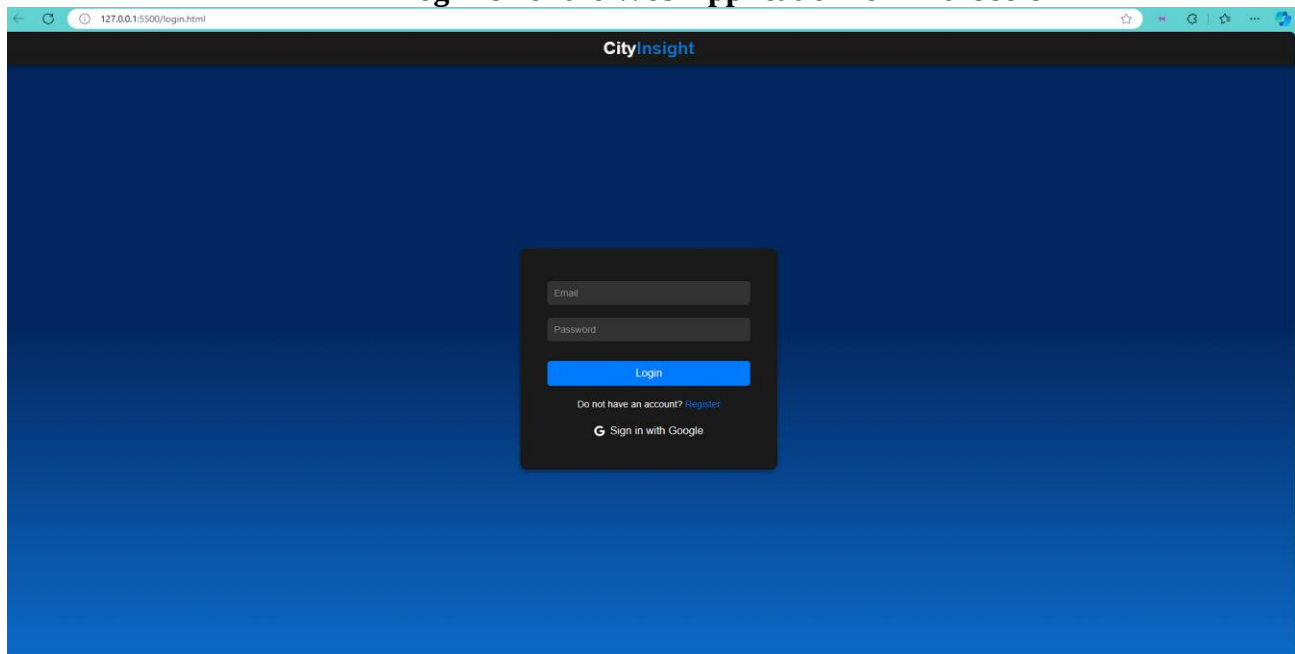


Plate 15 shows the Login UI of the web application. The registered Email Address and Password of the end users can be used in this page to be able to login within the app.

Plate # 16
Register UI of the Web Application for End Users

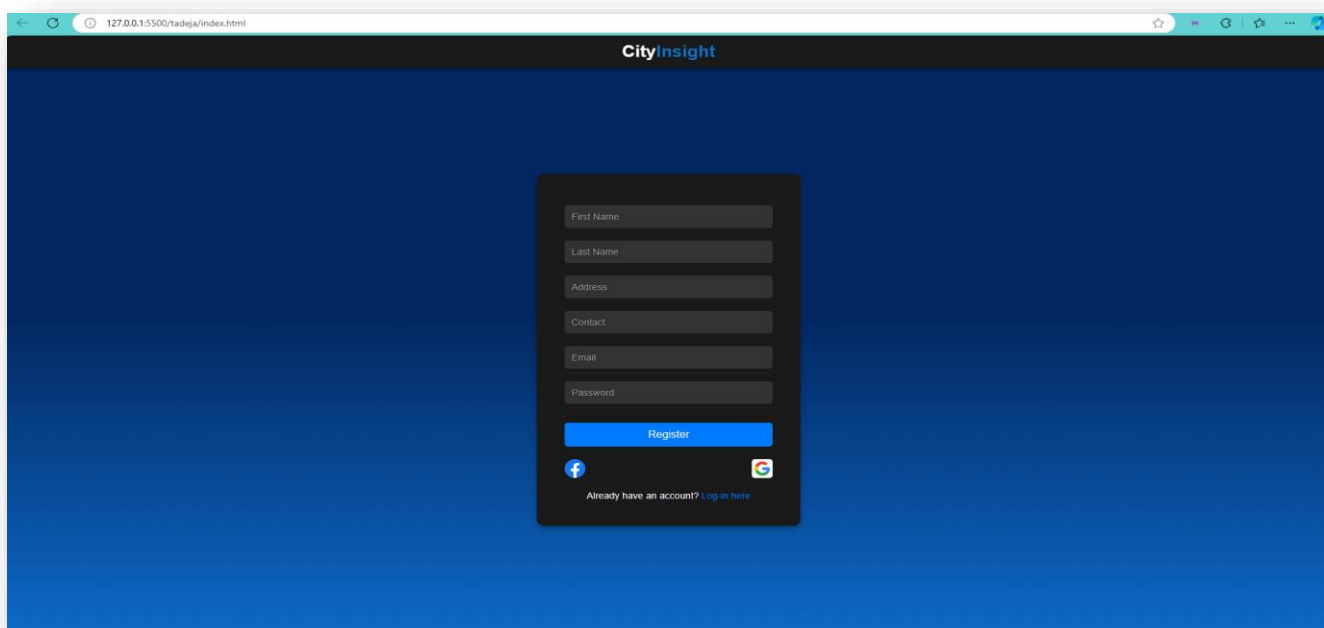




Plate 16 shows the Register UI of the web application. Register UI includes the user's information to register.

Plate # 17
Profile Information of the Web Application for End Users

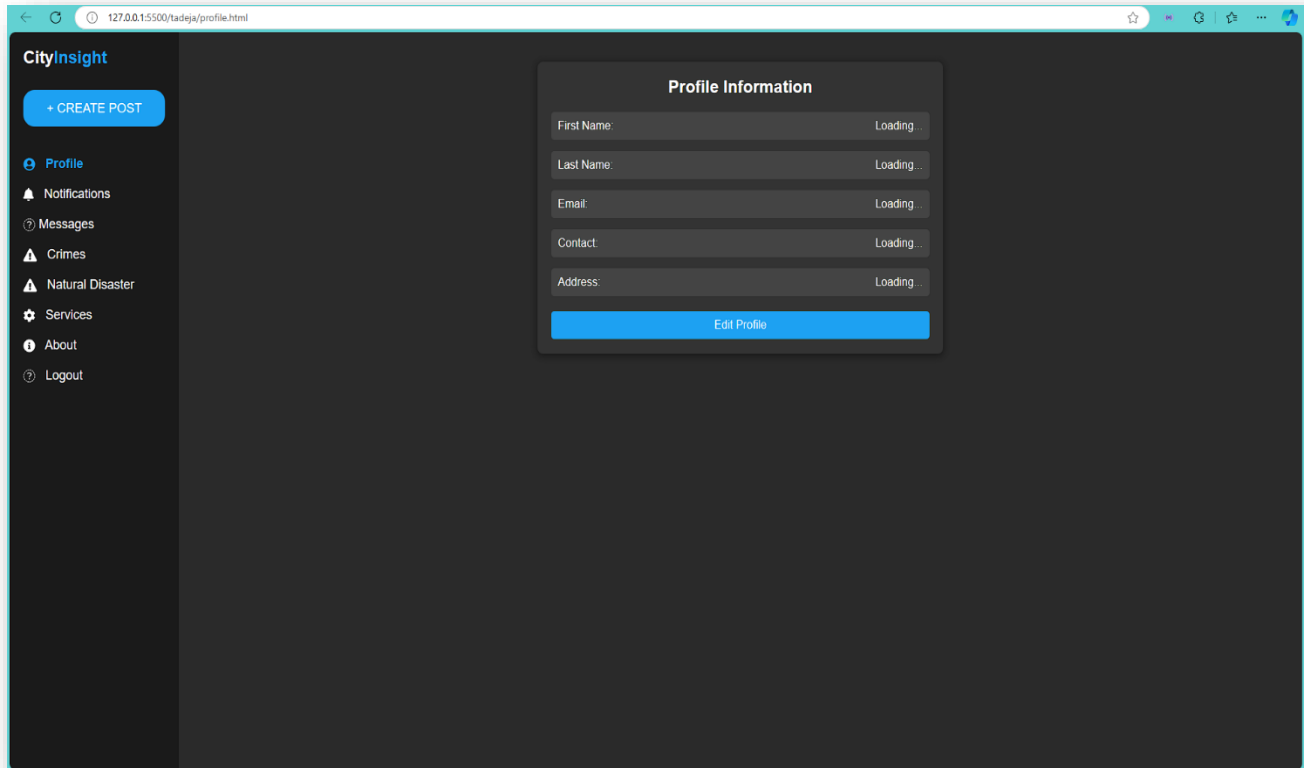


Plate 17 shows the Profile Information of the web application. Profile Information includes the user's information.

Plate # 18 Crime Mapping of the Web Application for End Users

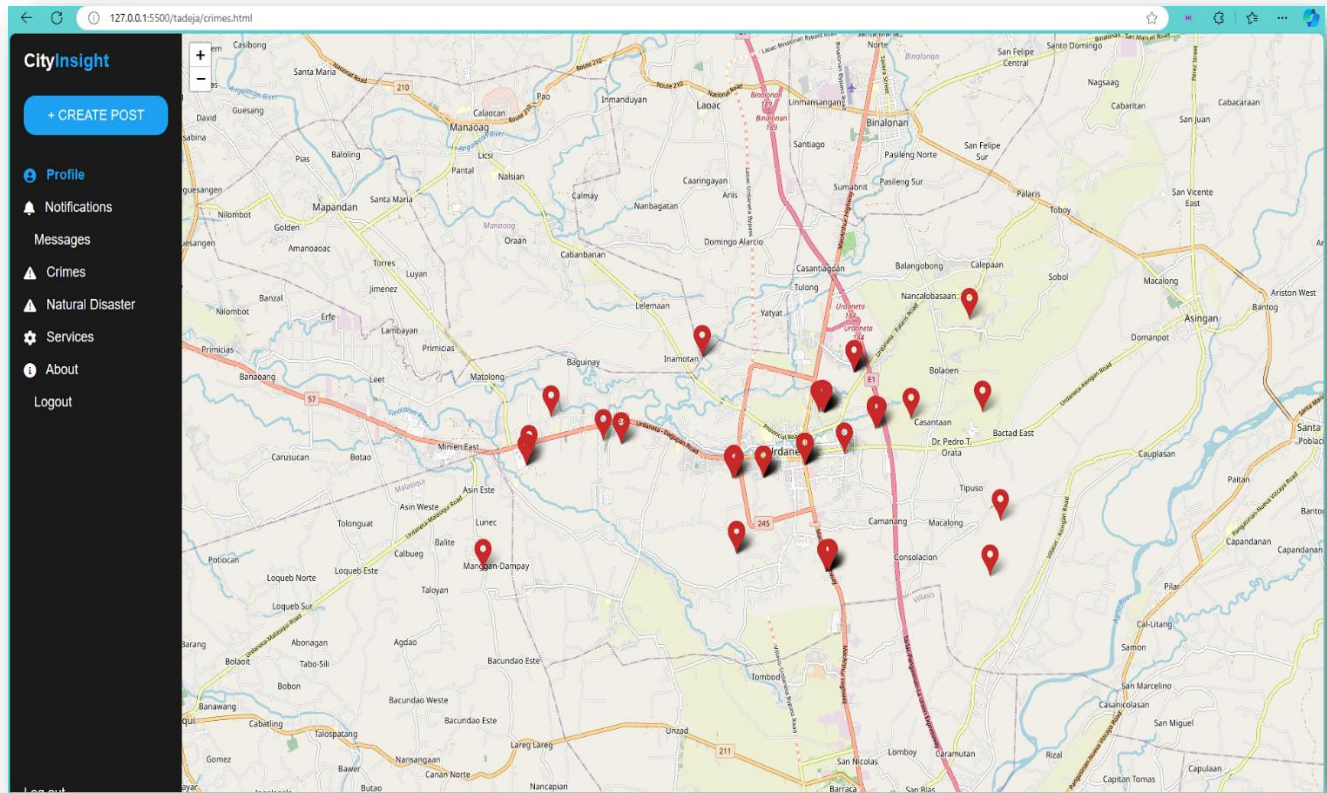


Plate 18 shows the Crime Mapping of the web application. Crime Mapping includes the map of Urdaneta City, Pangasinan and the location of Crime Incidents.

Plate # 19 Flood Mapping of the Web Application for End Users

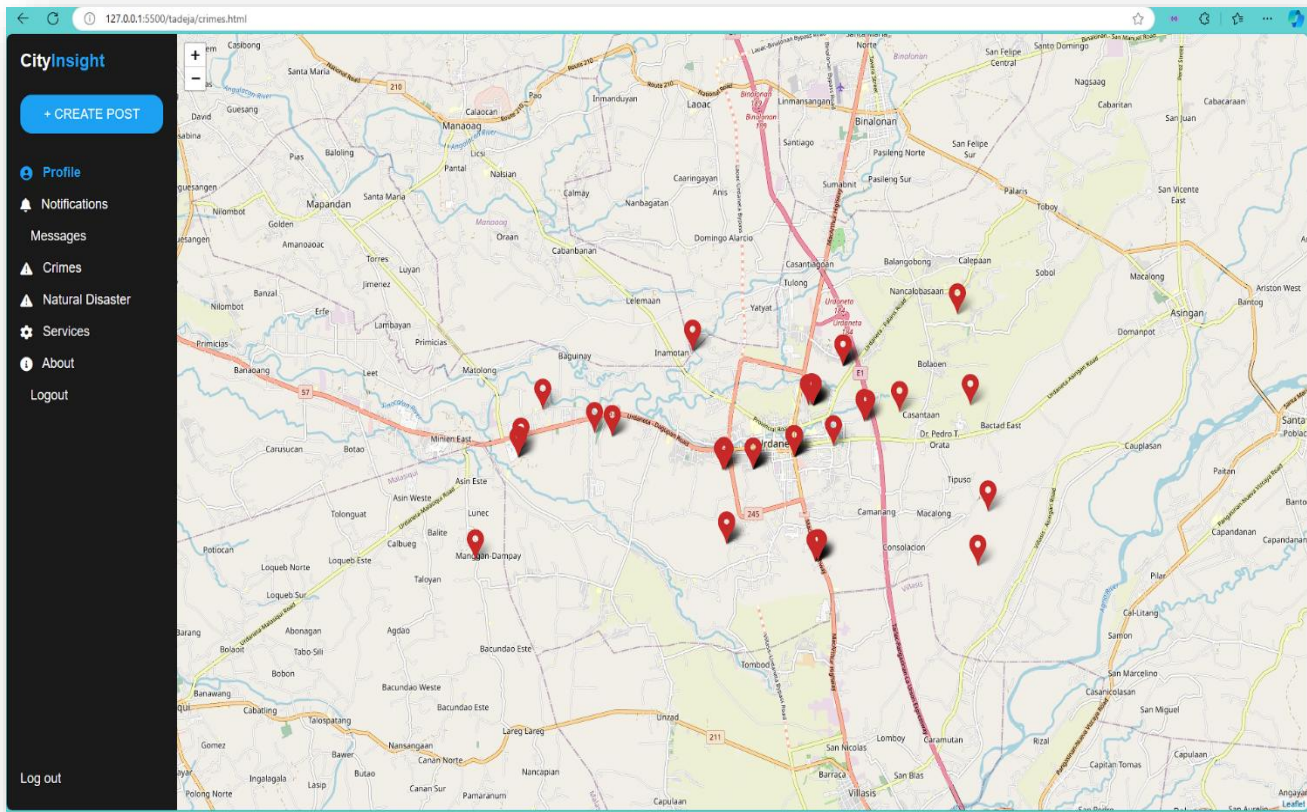


Plate 19 shows the Flood Mapping of the web application. Flood Mapping includes the map of Urdaneta City, Pangasinan and the location of Flood incidents.

Plate # 20
Crowdsourcing of the Web Application for End Users

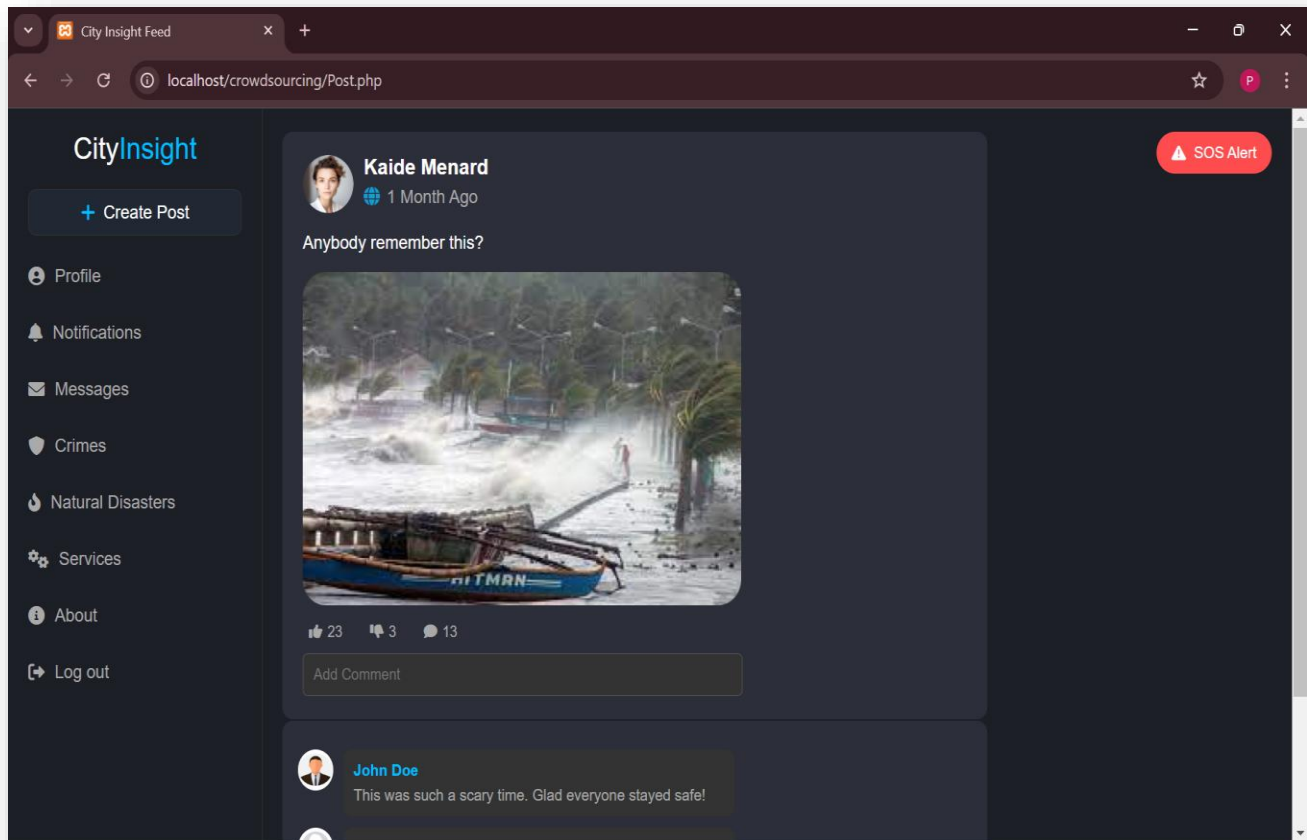


Plate 20 shows the Crowdsourcing of the Web application. Crowdsourcing include the posting of other users, and the other users can react and comment to the post.



Plate # 21
Create Post of the Web Application for End Users

A screenshot of a web browser displaying the "CityInsight Create Post" page. The browser's address bar shows "localhost/crowdsourcing/CreatePost.php". The page has a dark theme. On the left is a sidebar with the "CityInsight" logo and a "+ Create Post" button. Below the logo are menu items: Profile, Notifications, Messages, Crimes, Natural Disasters, Services, About, and Log out. The main content area is titled "Create Post" and contains a form with a "Title*" input field, a large text area for "Enter your post content here...", and a dashed box for media with the text "Drag and Drop images or videos or Browse". At the bottom of the form are "Save as Draft" and "Post" buttons. A red "SOS Alert" button is in the top right corner.

Plate 21 shows the Create Post of the Web application. The Create Post includes the posting of images or videos of crime or flood incidents.



Plate # 22
Chat UI of the Web Application for End Users

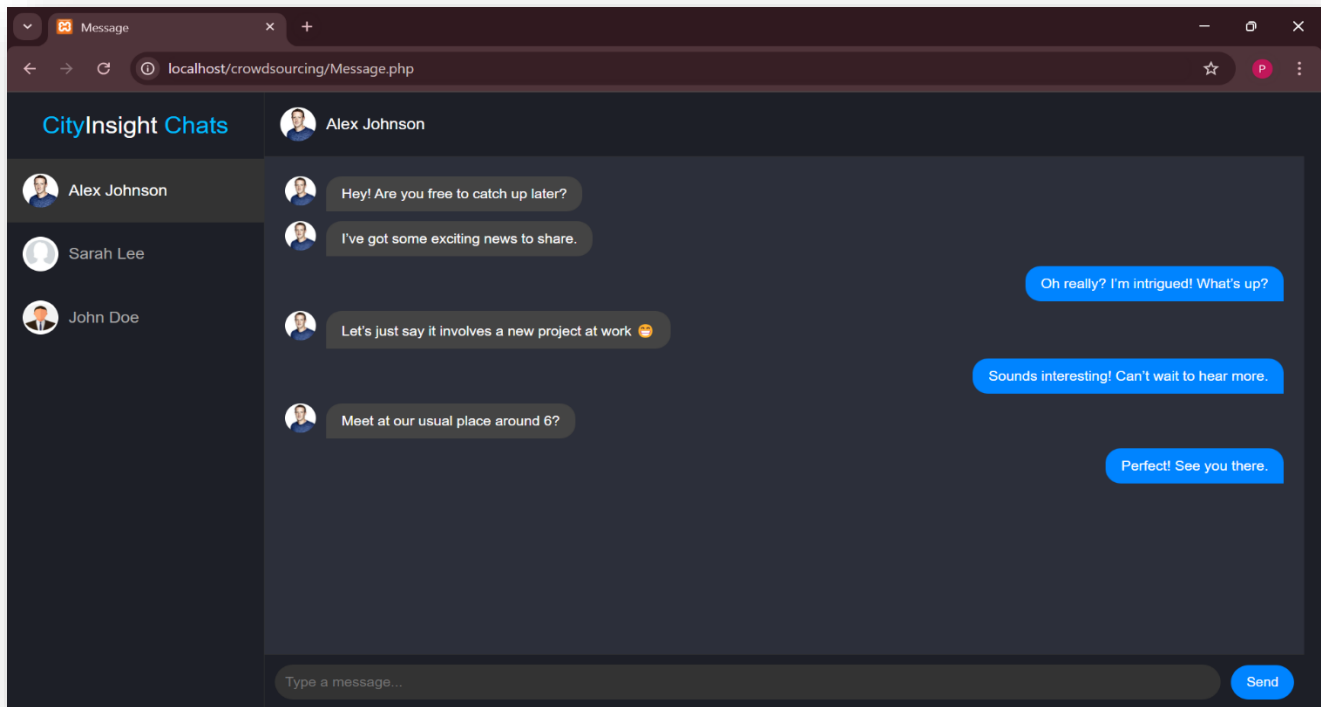


Plate 22 shows the Chat UI of the Web application. The Chat UI includes the chatting of the users.



ADMIN SIDE

Plate # 23

Adding of Crime Incidents of the users to the Super Admin

127.0.0.1:5500/admin/crimesuit.html

Google Maps - Urdaneta City, Pangasinan

Retam Kaide

- Dashboard
- Users
- Barangays
- Crime Reports
- Map Crime
- Map Flood
- SOS Alerts
- Archived Data
- Notifications
- Logout

Tag Location for crime reports.

Crime Type

Description

Date Tagged

Latitude

Longitude

Address

Add Crime

Plate 23 shows the Adding of Crimes Incidents of the Super Admin. The adding crimes incidents include the crime type, description, date tagged, latitude and longitude, and the address.



Plate # 24
Adding of Flood Incidents of the users to the Super Admin

The screenshot displays a web application interface for managing flood incidents. On the left is a dark sidebar with a user profile for 'Retsam Kalde' and a menu with options: Dashboard, Users, Barangays, Crime Reports, Map Crime, Map Flood (selected), SOS Alerts, Archived Data, Notifications, and Logout. The main area features a Google Map of Urdaneta City, Pangasinan, with various barangays labeled. Below the map is a form titled 'Tag Location for flood reports.' with input fields for Description, Date Tagged, Latitude, Longitude, and Address. A blue 'Add Crime' button is at the bottom of the form.

Plate 24 shows the Adding of Floods Incidents of the Super Admin. The adding floods incidents include the crime type, description, date tagged, latitude and longitude, and the address.



Plate # 25 SOS Request of the users to the Super Admin

Name	Latitude	Longitude
Christian Kaide	13.32323232323	14.32323232323
Christian Kaide	14.43434342343	14.43434342343

Plate 25 shows the SOS Request of the Super Admin. The SOS request allows the admin to see quickly the report emergencies or request assistance and streamline the process for immediate response and support.



Plate # 26
Barangay List of the Super Admin

The screenshot displays a web application interface for managing barangays in Urdaneta City. The sidebar on the left contains a user profile for 'Retsam Kalde' and a list of navigation items: Dashboard, Users, Barangays, Crime Reports, Map Crime, Map Flood, SOS Alerts, Archived Data, Notifications, and Logout. The main content area is titled 'Barangay in Urdaneta City' and features a table with the following data:

Barangay Name
Anonas
Bactad East
Bayaaos
Balaoen
Cabaruan
Cabuloan
Camanang
Camantiles
Casantaan
Catablan
Cayambanan
Consolacion
Dilan-Paurido
Labit Proper
Labit West
Mabanogbog
Macalong

Plate 26 shows the Barangay List of the Super Admin. The barangay list allows the admin to see the list of barangays in Urdaneta City.



Plate # 27
Dashboard of the Super Admin

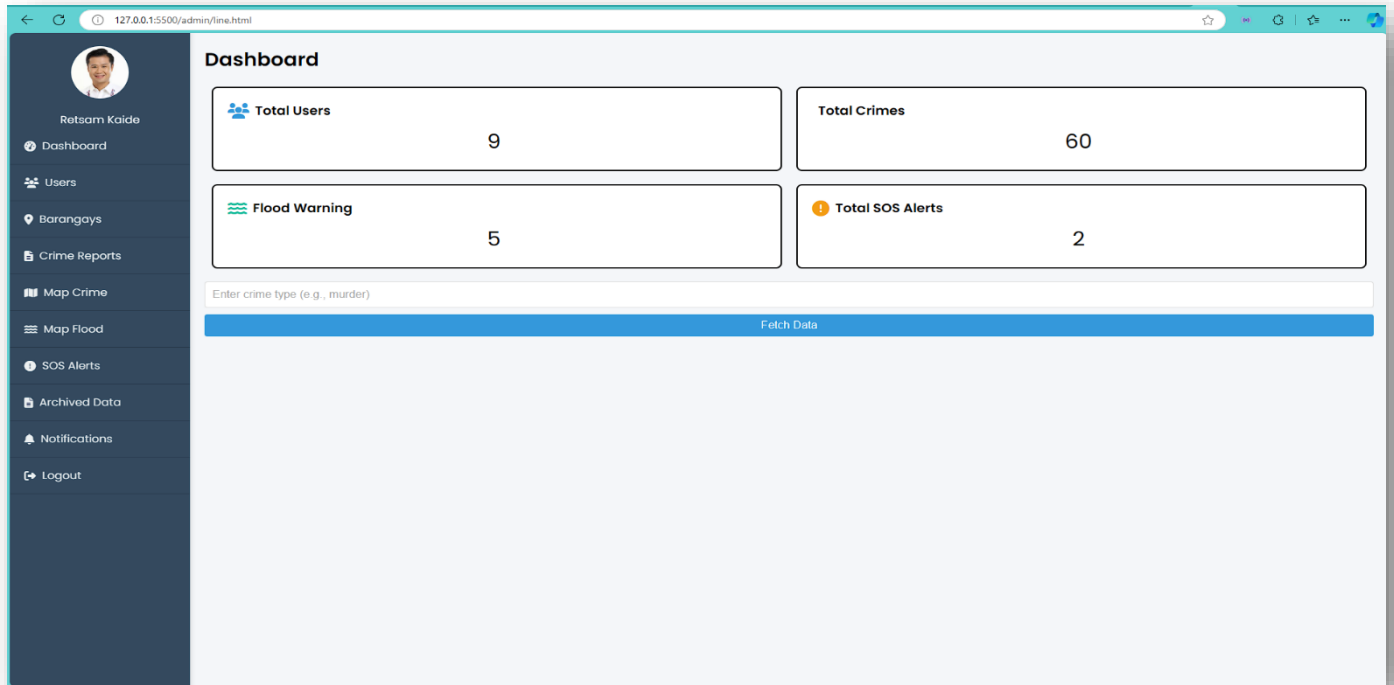


Plate 27 shows the Dashboard of the Super Admin. The Dashboard allows the admin to see the list of total users, total crimes, flood warnings and the total of SOS alerts.



Plate # 28
Users Table of the Registered Users

First Name	Last Name	Address	Contact	Email	Actions
Mark	Caps	86 Zamora Street	09090909090	mark@gmail.com	Delete
adw	qweq	qweq	123123134	noahqt250@gmail.com	Delete
qweqw	qweqw	qweqweqw	342342342	noahqt250@gmail.com	Delete
admin	admin	86 Zamora Street	09090909090	admin@gmail.com	Delete
qweqw	qweqwe	qweqwe	1231542344523	noahqt250@gmail.com	Delete
Police	Station	Urdaneta City	09917371829	police@gmail.com	Delete
ado	ado	japan	09874472574	luizaldrin250@gmail.com	Delete
qweqe	qweqwe	qweqeq	123134235	noahqt250@gmail.com	Delete
Christian	Tense	Imbalbalatong	09151581234	tencesen.69@gmail.com	Delete

Plate 28 shows the Users Table of the Registered Users. The user's tables allow the admin to see the list of total registered users and the admin can delete the users log in information.



Plate # 29

Adding the Crime Incidents of the Police

Google Maps - Urdaneta City, Pangasinan

Tag Location for crime reports.

Crime Type

Description

Date Tagged

Latitude

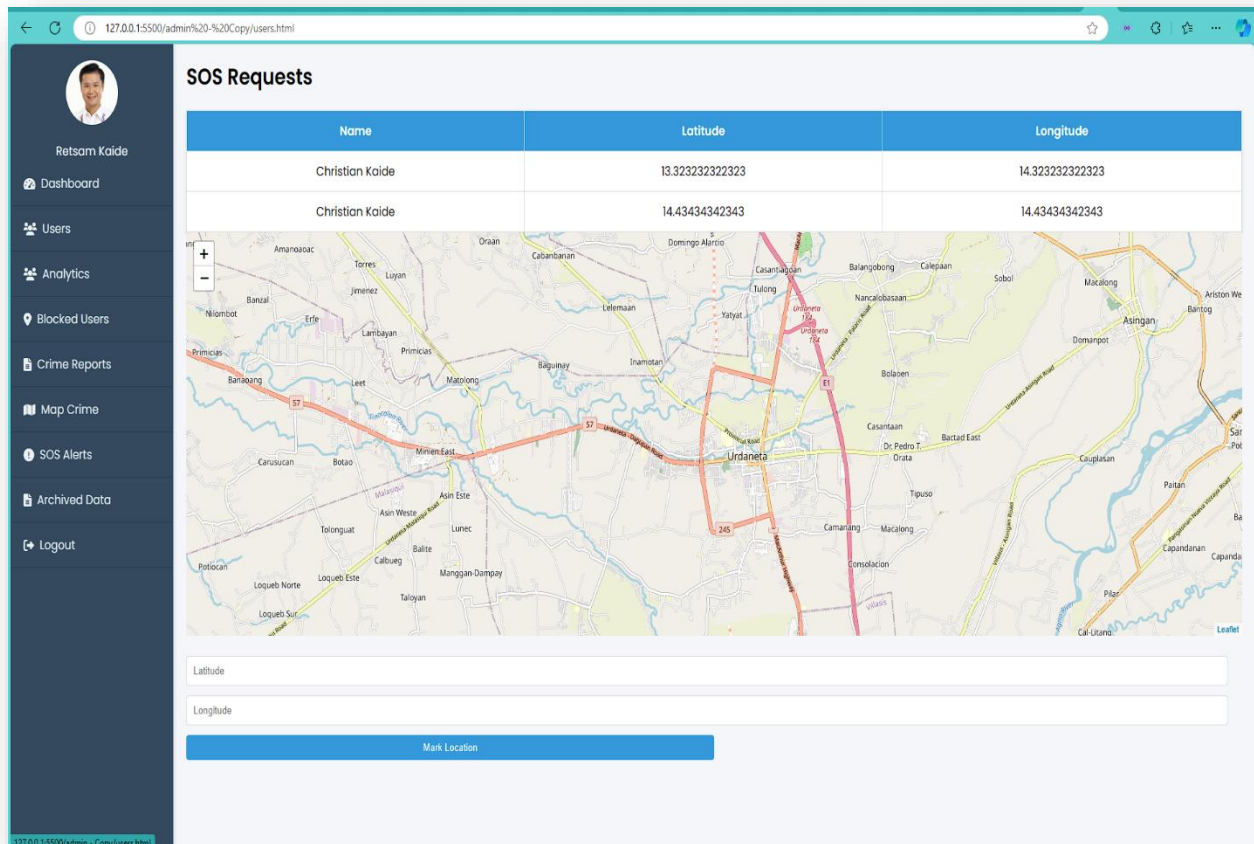
Longitude

Address

Add Crime

Plate 29 shows the Adding of Crimes Incidents of the Police. The adding crimes incidents include the crime type, description, date tagged, latitude and longitude, and the address.

Plate # 30
SOS Request of the users to the Police



Name	Latitude	Longitude
Christian Kaide	13.32323232323	14.32323232323
Christian Kaide	14.43434342343	14.43434342343

Plate 30 shows the SOS Request of the users to the police. The SOS request allows the police to see quickly the report emergencies or request assistance and streamline the process for immediate response and support.



Plate # 31
Crime Data of the Police

Barangay	Date	Latitude	Longitude	Type	Actions
Nancayasan	Timestamp(seconds=1727239500, nanoseconds=717000000)	15.950927329047976	120.57703918340204	THEFT - RPC Art. 308	Archive Delete
Catablan	Timestamp(seconds=1717513200, nanoseconds=907000000)	15.97474593452935	120.50209600035298	ANTI-VIOLENCE AGAINST WOMEN AND THEIR CHILDREN ACT OF 2004 - RA 9262	Archive Delete
Nancamaliran West	Timestamp(seconds=1717891200, nanoseconds=955000000)	15.970125816627615	120.55370606586888	GRAVE THREATS - RPC Art. 282	Archive Delete
Consolacion	Timestamp(seconds=1718089200, nanoseconds=636000000)	15.949782588277088	120.61730331260866	ANTI-VIOLENCE AGAINST WOMEN AND THEIR CHILDREN ACT OF 2004 - RA 9262	Archive Delete
Camantiles	Timestamp(seconds=1721314200, nanoseconds=412000000)	15.99543562477731	120.54533981624651	COMPREHENSIVE DANGEROUS DRUGS ACT OF 2002 - POSSESSION OF DANGEROUS DRUGS - RA 9165 Article II Section II	Archive Delete
Nancalobasaan	Timestamp(seconds=1718064000, nanoseconds=891000000)	16.003294831002254	120.61222362219634	ANTI-VIOLENCE AGAINST WOMEN AND THEIR CHILDREN ACT OF 2004 - RA 9262	Archive Delete
Dilan Paurido	Timestamp(seconds=1718181600, nanoseconds=702000000)	15.980549541049493	120.58868582561902	COMPREHENSIVE DANGEROUS DRUGS ACT OF 2002 - SALE, TRADING, ADMINISTRATION, DISPENSATION, DELIVERY, DISTRIBUTION AND TRANSPORTATION OF DANGEROUS DRUGS - RA 9165 Article II Section 5 para 1	Archive Delete
Nancalamiran East	Timestamp(seconds=1726647900, nanoseconds=932000000)	15.9702800726328	120.55303790844809	MURDER - RPC Art. 248	Archive Delete
Anonas	Timestamp(seconds=1727046000, nanoseconds=693000000)	15.992280251585303	120.58329902994538	RECKLESS IMPRUDENCE RESULTING TO PHYSICAL INJURY - RPC Art 365	Archive Delete
Nancalamiran	Timestamp(seconds=1727084400, nanoseconds=932000000)	15.970444401285327	120.56073576168335	MURDER - RPC Art. 248	Archive Delete

Plate 31 shows the Crime reports of the users to the police. Crime data management allows the police to archive and delete the reports.

Plate # 32 Analytics of Crime Types in Each Barangay

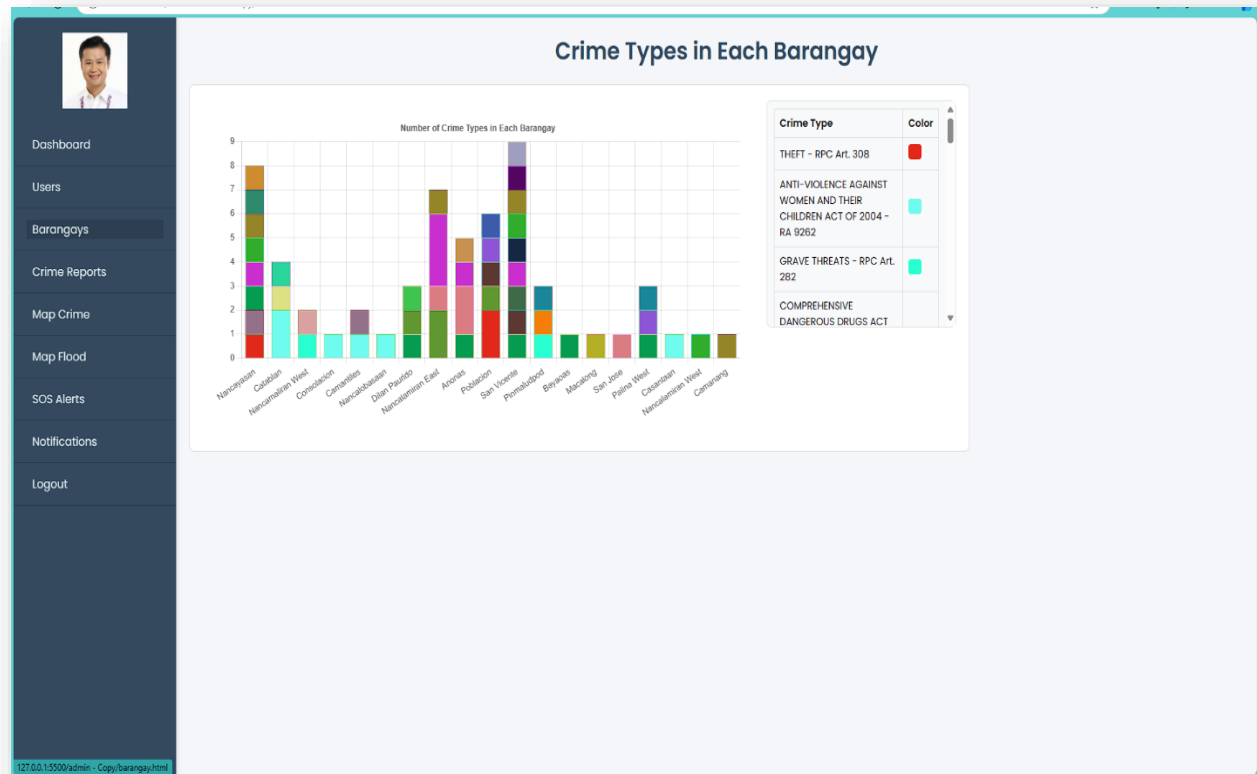


Plate 32 shows the Bar Graph of the Crime types in each barangay. The Analytics allows the police see the graph of per barangay and the number of crimes.



Plate # 33
Registered Users of the Mobile and Web Application

First Name	Last Name	Address	Contact	Email	Actions
Mark	Caps	86 Zamora Street	09090909090	mark@gmail.com	Unlocked
adw	qweq	qweq	123123134	noahqt250@gmail.com	Unlocked
qweqw	qweqw	qweqweqw	342342342	noahqt250@gmail.com	Unlocked
admin	admin	86 Zamora Street	09090909090	admin@gmail.com	Unlocked
qweqw	qweqwe	qweqwe	1231542344523	noahqt250@gmail.com	Unlocked
Police	Station	Urdaneta City	09917371829	police@gmail.com	Unlocked
ado	ado	japan	09874472574	luizaldrin250@gmail.com	Unlocked
qweqe	qweqwe	qweqeq	123134235	noahqt250@gmail.com	Unlocked
Christian	Tense	Imbalalatong	09151581234	tencesen.69@gmail.com	Unlocked

Plate 33 shows the Registered users among the Mobile and Web Application. The police allow blocked or unblocked the users.



Plate # 34
Dashboard of the Police

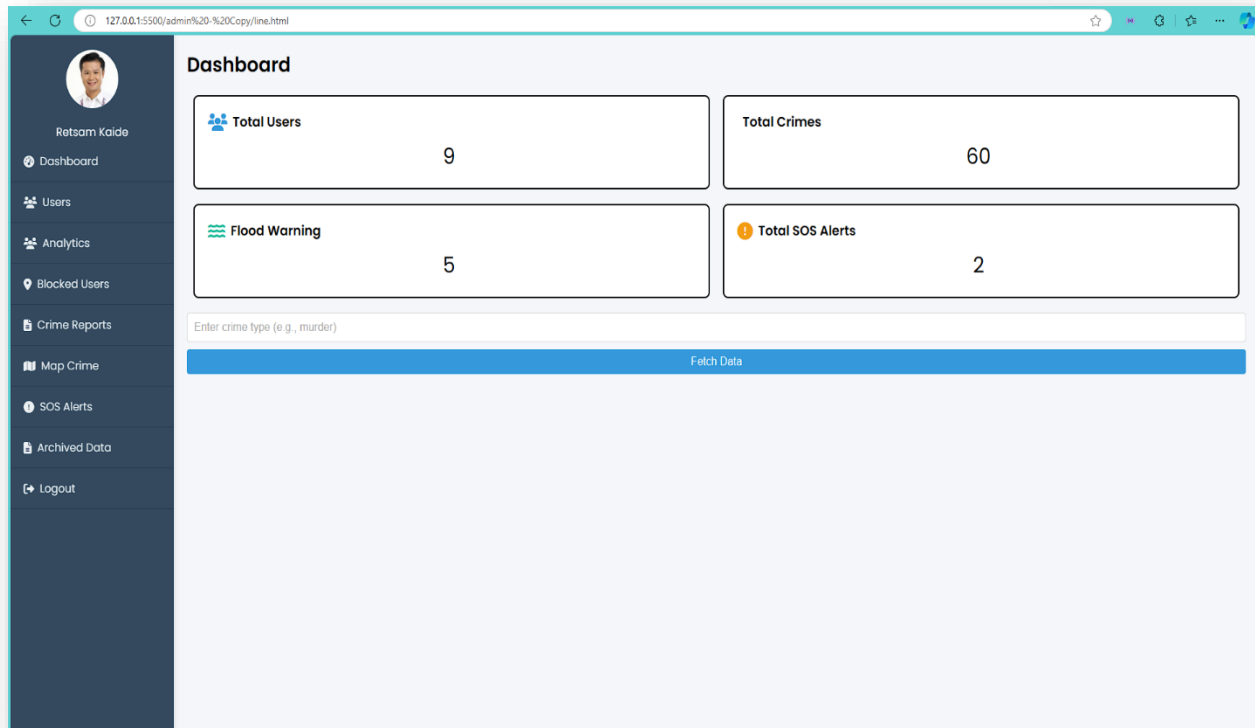


Plate 34 shows the Dashboard of the Police. The Dashboard allows the admin to see the list of total users, total crimes, flood warnings and the total of SOS alerts.



Plate # 35
Registered Users of the Mobile and Web Application

 Retsam Kaide Dashboard Users Analytics Blocked Users Crime Reports Map Crime SOS Alerts Archived Data Logout	Registered Users					
	First Name	Last Name	Address	Contact	Email	Actions
	Mark	Caps	86 Zamora Street	09090909090	mark@gmail.com	Delete
	adw	qweq	qweq	123123134	noahqt250@gmail.com	Delete
	qweqw	qweqw	qweqweqw	342342342	noahqt250@gmail.com	Delete
	admin	admin	86 Zamora Street	09090909090	admin@gmail.com	Delete
	qweqw	qweqwe	qweqwe	1231542344523	noahqt250@gmail.com	Delete
	Police	Station	Urdaneta City	09917371829	police@gmail.com	Delete
	ado	ado	japan	09874472574	lulzaldin250@gmail.com	Delete
	qweqe	qweqwe	qweqeq	123134235	noahqt250@gmail.com	Delete
	Christian	Tense	Imbalbalatong	09151581234	tencesen.69@gmail.com	Delete

Plate 35 shows the Users Table of the Registered Users. The user's tables allow the police to see the list of total registered users and the police can delete the users log in information.



Plate # 36 Adding the Flood Incidents of the NDRRMC

Google Maps - Urdaneta City, Pangasinan

Tag Location for crime reports.

Crime Type

Description

Date Tagged

Latitude

Longitude

Address

Add Crime

Plate 36 shows the Adding of Floods Incidents of the NDRRMC. The adding flood incidents include the flood type, description, date tagged, latitude and longitude, and the address.



Plate # 37
SOS Request of the users to the NDRRMC

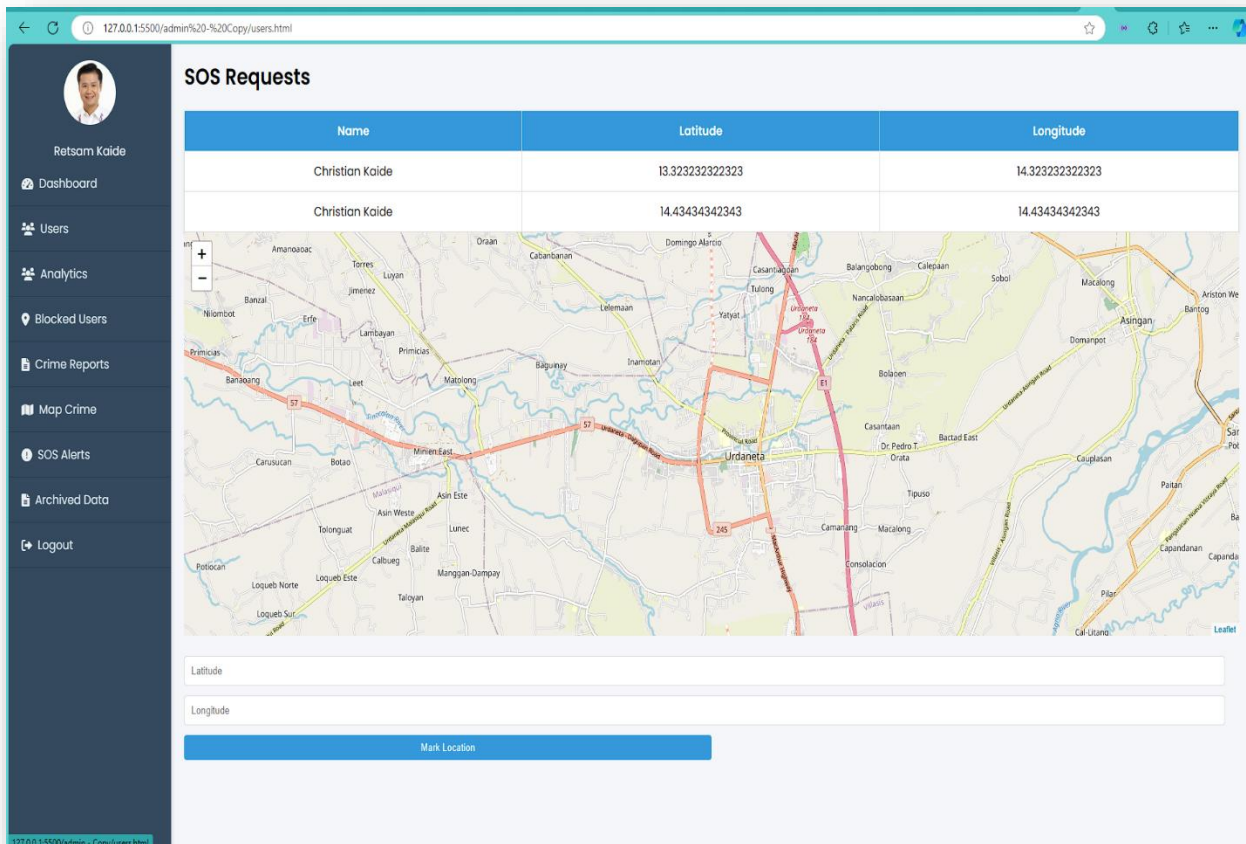


Plate 37 shows the SOS Request of the users to the NDRRMC. The SOS request allows the NDRRMC to see quickly the report emergencies or request assistance and streamline the process for immediate response and support.



Plate # 38
Dashboard of the NDRRMC

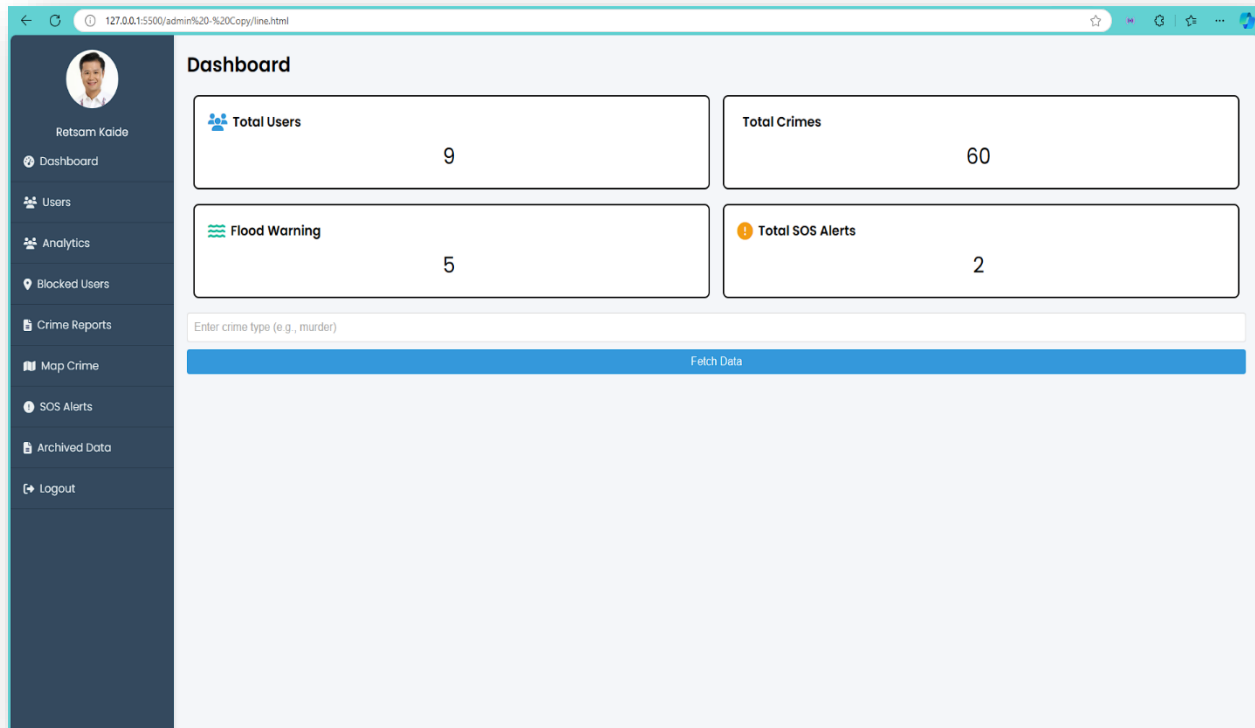


Plate 38 shows the Dashboard of the CDRRMC. The Dashboard allows the NDRRMC to see the list of total users, total crimes, flood warnings and the total of SOS alerts.



Plate # 39
Barangay List of the CDRRMC

The screenshot displays a web application interface for the CDRRMC. On the left is a dark sidebar with a user profile for 'Retsam Kalde' and a list of navigation items: Dashboard, Users, Barangays, Crime Reports, Map Crime, Map Flood, SOS Alerts, Archived Data, Notifications, and Logout. The main content area is titled 'Barangay in Urdaneta City' and contains a table with the following data:

Barangay Name
Anonas
Bactad East
Bayabas
Balaoen
Cabaruan
Cabuloan
Camanang
Camantiles
Casantaan
Catablan
Cayambanan
Consolacion
Dilan-Paurido
Labit Proper
Labit West
Mabanogbog
Macalong

Plate 39 shows the Barangay List of the CDRRMC. The barangay list allows the CDRRMC to see the list of barangays in Urdaneta City.



Usability of CityInsight: An Urban Safety and Emergency Response Application for Urdaneta City

Before a mobile app and web app is made available to the public, it is crucial to assess the app's usability. One approach to perform usability research is to utilize a usability questionnaire to gather input from the target audience. Software Usability Measurement Inventory (SUMI) was employed by the developers to gauge the usability level of CityInsight among possible end users.

Table 4. Software Usability Measurement Inventory (SUMI) Results for CityInsight: Web and Mobile-based Personal Safety Companion

Statement	Mean Score	Efficiency	
		Descriptive Equivalent	Descriptive Interpretation
1. It takes too long to learn the software functions.	2.50	Moderately Agree	Usable
2. I would recommend this software to my colleagues.	3.73	Agree	Usable
3. Learning to operate this software initially is full of problems.	2.47	Disagree	Not Usable
4. The software has a steep learning curve.	2.72	Moderately Agree	Usable
5. I find the software difficult to navigate.	2.00	Disagree	Not Usable
6. I would suggest this software to others in my field.	3.52	Agree	Usable
7. The initial setup of the software is challenging.	2.58	Moderately Agree	Usable
8. Mastering the software's features requires significant time and effort.	2.95	Moderately Agree	Usable
9. I feel confident in recommending this software to peers.	3.79	Agree	Usable
10. The software's functionality is hard to	2.94	Moderately Agree	Usable



grasp at first.			
Affect			
Statement	Mean Score	Descriptive Equivalent	Descriptive Interpretation
1. There is never enough information on the screen when it's needed.	2.37	Disagree	Not Usable
2. I feel safer if I use only a few familiar functions.	3.21	Moderately Agree	Usable
3. The software often lacks sufficient details when I need them.	2.37	Disagree	Not Usable
4. I prefer to use only the functions I'm familiar with for better security.	3.10	Moderately Agree	Usable
5. The information displayed on the screen is frequently inadequate.	2.37	Disagree	Not Usable
6. Using only a few well-known functions makes me feel more secure.	2.95	Moderately Agree	Usable
7. There is often a shortage of relevant information on the screen.	2.31	Disagree	Not Usable
8. I feel more comfortable sticking to a limited set of features.	2.68	Moderately Agree	Usable
9. The screen often doesn't provide enough context when needed.	2.47	Disagree	Not Usable
Control			
Statement	Mean Score	Descriptive Equivalent	Descriptive Interpretation
1. The organization of the menus seems quite logical.	3.47	Moderately Agree	Usable
2. There are too many steps required to get something to work.	2.68	Moderately Agree	Usable
3. I will never learn to use all that is offered in this software.	2.31	Disagree	Not Usable



4. The menu structure is intuitive and easy to follow.	4.00	Agree	Usable
5. It takes too many steps to accomplish simple tasks with this software.	2.53	Moderately Agree	Usable
6. I doubt I'll ever master all the features available in this software.	3.73	Agree	Usable
7. The way the menus are arranged makes sense to me.	3.73	Agree	Usable
8. The number of steps needed to achieve a goal with this software feels excessive.	3.53	Agree	Usable

Helpfulness			
Statement	Mean Score	Descriptive Equivalent	Descriptive Interpretation
1. There is too much to read before you can use the software.	2.84	Moderately Agree	Usable
2. The software has helped me overcome any problems I have had in using it.	3.63	Agree	Usable
3. I keep having to go back to look at the guides.	2.57	Moderately Agree	Usable
4. The amount of documentation required before using the software is overwhelming.	2.47	Disagree	Not Usable
5. The software has effectively resolved any issues I've encountered.	3.73	Agree	Usable
6. I find myself frequently referring to the user guides.	2.89	Moderately Agree	Usable
7. The software requires extensive reading before I can start using it effectively.	2.57	Moderately Agree	Usable
8. Any problems I face with the software are	3.94	Agree	Usable



easily resolved through its features.			
9. I often need to consult the guides to understand how to use the software.	3.80	Agree	Usable
10. The initial learning curve involves too much reading and preparation.	3.90	Agree	Usable

Table 5. Overall Result of Usability Test

Dimension	Mean Score	Descriptive Equivalent	Descriptive Interpretation
Efficiency	3.00	Moderately Agree	Usable
Affect	2.88	Moderately Agree	Usable
Control	3.34	Agree	Usable
Helpfulness	3.15	Agree	Usable
Weighted Mean	3.01	Moderately Agree	Usable

Table 5 shows the summary table highlights the usability evaluation of CityInsight based on four dimensions: Efficiency, Affect, Control, and Helpfulness. The software received an overall weighted mean of 3.01, with users generally rating it as "Usable." While Control received the highest score (3.34), indicating strong agreement on menu organization and ease of navigation, Efficiency and Affect scored moderately, pointing to areas for improvement in learning and interface clarity. Helpfulness scored well, showing that guides and support effectively address user needs.



CHAPTER 5

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

In this chapter, the summary, conclusions, and recommendation provided by the developers are presented based on the objectives in the development of the study. The results led to the drawing of conclusions and to the arrival of recommendations.

Summary

The study titled "*CityInsight: Web and Mobile-based Personal Safety Companion*" was developed to investigate how a technology-driven platform can enhance safety and emergency response efforts in urban environments. In response to the growing need for effective public safety measures, the project aimed to create a comprehensive mobile and web application that facilitates real-time reporting, access to emergency services, and up-to-date crime information for Urdaneta City residents, commuters, and local authorities.

CityInsight is designed to bridge communication between users and emergency responders by providing essential features such as crime mapping, incident geotagging, emergency alerts, and local safety tips. This study focused on identifying the specific requirements of various user groups—including the Philippine National Police (PNP), the National Disaster Risk Reduction and Management Council (NDRRMC), residents, students, and travelers to ensure the platform's functionality meets the community's safety needs.

Using SUMI Methodology, the developers created an iterative process for building, testing, and refining the application based on user feedback. A structured survey was distributed among Urdaneta City residents and relevant stakeholders to assess their perceptions of the application's usability, relevance, and efficiency in promoting public safety. The survey data provided insights



into user preferences, including highly valued features like emergency hotlines, safety tips, crowdsourced crime reporting, and historical crime data.

The goal of the CityInsight project is to empower users with tools to navigate urban risks, report incidents quickly, and access reliable information that can contribute to safer communities. Through a user-centered design and ongoing improvements, CityInsight seeks to foster a proactive and informed community, bridging the gap between technology and public safety in Urdaneta City.

Conclusion

Based on the results of the conducted research, the following conclusions are made by the researchers:

1. The developers concluded that various user groups, including PNP, NDRRMC, Urdaneta City residents, and students, are prepared to use the CityInsight application after undergoing a comprehensive assessment and feedback-gathering process from these stakeholders.
2. The need for quick access to emergency alerts, crime mapping, and geotagging features was evident, as these functionalities were highly prioritized by the respondents. The existing methods for crime and incident awareness rely heavily on traditional forms of communication, indicating a gap that CityInsight aims to address through a more accessible and digital approach.
3. Feedback from residents, travelers, and commuters in Urdaneta City indicated a positive response toward the usability and features of CityInsight. The survey checklist revealed a strong interest in functions like real-time emergency hotlines, community-based reporting, and historical crime logs, which will enhance users' awareness and security.



4. The developers were able to utilize agile practices effectively through the SUMI methodology, which helped streamline the development process and address the unique needs of each stakeholder group involved in the project.
5. The system usability measurement inventory test provided a positive review, affirming that CityInsight meets the necessary standards for functionality, learnability, and ease of navigation, ensuring that the application will be user-friendly and efficient for all target users.

Recommendations

Based on the results and conclusions made in the study, the following recommendations are hereby given:

1. The appropriate evaluation has been completed, and the created CityInsight application is now ready for usage by both residents and officials in Urdaneta City.
2. It is recommended that local emergency response teams, such as the PNP and NDRRMC, utilize the app's features, such as emergency alerts and crime mapping, to enhance their operational efficiency and community engagement in real-time scenarios.
3. The CityInsight application would greatly benefit residents, travelers, and students by providing quick access to critical information like safety tips and geotagged incident locations, thus increasing their sense of security and awareness.
4. The developers recommend continued use of the SUMI methodology for future updates to CityInsight, as it allows for adaptable planning, iterative improvements, and ongoing input, ensuring the application continues to meet the evolving needs of the community.



5. It is advised that the City Government of Urdaneta and surrounding municipalities in Pangasinan adopt and implement the CityInsight application to improve local safety, awareness, and emergency preparedness across a wider region



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[Shabu/publication/324164102_Natural_Disasters_and_Crime_Incidence_A_Case_of_2012_Flooding_in_Benue_State_Nigeria/links/5ac27c63aca27222c75ce9c6/Natural-Disasters-and-Crime-Incidence-A-Case-of-2012-Flooding-in-Benue-State-Nigeria.pdf](https://www.researchgate.net/profile/Terwase-Shabu/publication/324164102_Natural_Disasters_and_Crime_Incidence_A_Case_of_2012_Flooding_in_Benue_State_Nigeria/links/5ac27c63aca27222c75ce9c6/Natural-Disasters-and-Crime-Incidence-A-Case-of-2012-Flooding-in-Benue-State-Nigeria.pdf)

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APPENDIX A

Structured interview for The Police-in-Charge of PNP Urdaneta respondent to determine the protocols, processes, and procedures required to properly disperse the crime incidents.

Structured Interview for Crime Incident Dispersal:

I. General Information

1. Can you describe your role in handling crime incidents within PNP Urdaneta/NDRRMC?
 - What are your primary responsibilities?
 - How long have you been in this role?
2. What are the most common types of crime incidents you deal with?
 - Do the types of incidents vary by season or other factors?

II. Protocols for Incident Reporting

1. What is the standard protocol for reporting a crime incident?
 - How is the information initially received (e.g., via 911, direct report, etc.)?
 - Who is responsible for receiving the initial report?
2. What steps are taken once a crime incident is reported?
 - Who gets notified first, and how?
 - Are there specific guidelines for prioritizing incidents?
3. How are crime scenes managed?
 - What procedures are in place for securing a crime scene?
 - How is evidence handled and documented?



III. Incident Response Procedures

1. What are the established procedures for responding to different types of crime incidents (e.g., theft, homicide, terrorism)?
 - How do you assess the intensity and respond accordingly?
2. How is the deployment of personnel decided during an incident?
 - What factors are considered?
3. What tools and technologies are used to support incident response?
 - Are there any specific software or communication devices used?

IV. Post-Incident Procedures

1. What processes are in place for debriefing after an incident?
 - Who is involved in the debriefing?
2. How are reports generated and reviewed post-incident?
 - What information is typically included in these reports?
3. How is feedback from past incidents used to improve future responses?

V. Conclusion

4. Is there anything else you would like to add regarding the crime incident dispersal protocols, processes, and procedures?
5. Would you be willing to participate in future discussions or provide additional insights as we continue this review?



APPENDIX B

Structured Interview for Disaster Preparedness and Response:

II. Organizational Role and Responsibilities

1.Can you describe your role within the NDRRMC?

- What are your main responsibilities concerning disaster management?
- How long have you been in this position?

2.How does your role fit within the broader NDRRMC disaster response framework?

- What other departments or agencies do you collaborate with?

III. Preparedness Protocols

1.What are the key protocols in place for disaster preparedness?

- How do you ensure the community is prepared for potential disasters?
- What regular training or drills are conducted for personnel?

2.How is disaster risk assessed in different regions?

- What criteria are used to determine high-risk areas?
- How often are risk assessments updated?

IV. Incident Response Protocols

1.What is the standard procedure for responding to a disaster incident?

- How are initial alerts and warnings communicated to the public?



Bachelor of Science in Information Technology

- What steps are taken to activate emergency response teams?

2. How is the coordination between different agencies managed during a disaster?

- What is the chain of command during an incident?
- How is real-time information shared among agencies?

3. What tools and technologies are utilized in disaster response?

- Are there specific software systems or communication platforms in place?
- How do these tools aid in decision-making during an emergency?

IV. Post-Disaster Procedures

4. What are the steps involved in post-disaster recovery and evaluation?

- How is the effectiveness of the response assessed?
- What processes are in place for providing aid and resources to affected communities?

5. How are lessons learned from past disasters incorporated into future planning?

- What mechanisms are used for collecting feedback?
- How is this feedback used to improve future disaster responses?

V. Community Involvement and Awareness

1. How does the NDRRMC engage with local communities to enhance disaster preparedness?

- Are there programs or initiatives aimed at raising awareness?
- How is public input incorporated into disaster planning?

2. What roles do community leaders and organizations play in disaster response?



- How are they trained or prepared to assist during an incident?

VI. Conclusion

1. Is there anything else you would like to add regarding disaster preparedness and response protocols?
2. Would you be willing to participate in future discussions or provide additional insights as we continue to review and enhance our disaster management strategies?



APPENDIX C

The survey checklist aims to determine the CityInsight user requirements for PNP, NDRRMC, Urdaneta City Residents, and Urdaneta City Students.

Instructions: These are some of the system's initial features. Choose the ones you believe will be most useful to you when using the suggested program by checking the appropriate box corresponding to your level of agreement. Use the following scale to indicate your response:

- **5 - Extremely Important**
- **4 - Very Important**
- **3 - Moderately Important**
- **2 - Slightly Important**
- **1 - Not Important**

Please feel free to give any comments or recommendations that will help us in determining what features to add or improve.

FEATURES	5	4	3	2	1
Crime Mapping: Visualize crime incidents on a map to identify hotspots and patterns in your area.					



Crowdsourcing: Gather community input to report and verify criminal activities or suspicious behavior.					
Emergency Alert: Receive real-time notifications about nearby emergencies or ongoing dangerous situations.					
Emergency Hotlines: Quick access to essential contact numbers for police, medical, and fire services.					
Posting Crime Scenes: Share and view images or videos of crime scenes to raise awareness and aid investigations.					
Geotagging: Attach location data to photos, videos, and reports to pinpoint the exact location of incidents.					
Safety Tips: Access helpful advice and best practices for staying safe in various situations					



Incident History Logs: Access a detailed history of crime reports and alerts in your area.					
Local Crime News: Stay updated with the latest crime news specific to your area.					
Crime Heatmaps: View color-coded maps highlighting areas with higher concentrations of crime.					

If you have any comments or suggestions, please write down below.



APPENDIX D

Instructions: This survey is designed to gather insights and feedback from residents of Urdaneta City, as well as travelers and commuters, including students. The checklist consists of a series of statements regarding your experiences, preferences, and opinions about certain features or aspects relevant to the CityInsight app and its usability, especially in the context of crime and safety awareness.

Please indicate the extent to which you agree or disagree with each statement by selecting one of the following options:

- **1 – Strongly Disagree:** You completely disagree with the statement, indicating it does not reflect your view or experience at all.
- **2 – Disagree:** You generally disagree with the statement, although you may find it partially applicable.
- **3 – Moderately Agree:** You somewhat agree with the statement, but with reservations; it partially reflects your view or experience.
- **4 – Agree:** You mostly agree with the statement, indicating it aligns well with your experience or opinion.
- **5 – Strongly Agree:** You completely agree with the statement, indicating it accurately reflects your view or experience.

Statement	1	2	3	4	5
EFFICIENCY					
It takes too long to learn the software functions.					
I would recommend this software to my colleagues.					
Learning to operate this software initially is full of problems.					



The software has a steep learning curve.					
I find the software difficult to navigate.					
I would suggest this software to others in my field.					
The initial setup of the software is challenging.					
Mastering the software's features requires significant time and effort.					
I feel confident in recommending this software to peers.					
The software's functionality is hard to grasp at first.					
AFFECT					
There is never enough information on the screen when it's needed.					
I feel safer if I use only a few familiar functions.					
The software often lacks sufficient details when I need them.					



I prefer to use only the functions I'm familiar with for better security.					
The information displayed on the screen is frequently inadequate.					
Using only a few well-known functions makes me feel more secure.					
There is often a shortage of relevant information on the screen.					
I feel more comfortable sticking to a limited set of features.					
The screen often doesn't provide enough context when needed.					
HELPFULNESS					
There is too much to read before you can use the software.					
The software has helped me overcome all the problems I have had with using it.					
I keep having to go back to look at the guides.					



The amount of documentation required before using the software is overwhelming.					
The software has effectively resolved any issues I've encountered.					
I find myself frequently referring to the user guides.					
There is too much to read before you can use the software.					
The software has helped me overcome any problems I have had in using it.					
I keep having to go back to look at the guides.					
The amount of documentation required before using the software is overwhelming.					
The software has effectively resolved any issues I've encountered.					
I find myself frequently referring to the user guides.					
The software requires extensive reading before I can start using it effectively.					
Any problems I face with the					



software are easily resolved through its features.					
I often need to consult the guides to understand how to use the software.					
The initial learning curve involves too much reading and preparation.					
CONTROL					
The organization of the menus seems quite logical.					
There are too many steps required to get something to work.					
I will never learn to use all that is offered in this software.					
The menu structure is intuitive and easy to follow.					
It takes too many steps to accomplish simple tasks with this software.					
I doubt I'll ever master all the features available in this software.					
The way the menus are arranged makes sense to me.					



The number of steps needed to achieve a goal with this software feels excessive.					
It seems unlikely that I'll become proficient in every aspect of this software.					
The organization of the software's options is user-friendly and logical.					
LEARNABILITY					
It is relatively easy to move from one part of a task to another.					
Getting data files in and out of the system is not easy.					
Either the amount or quality of the help information varies across					
Transitioning between different sections of a task is straightforward.					
Importing and exporting data files is challenging with this system.					
The help resources available vary in detail and usefulness.					



Navigating through various stages of a task is relatively simple.					
Managing data file transfers is cumbersome and problematic.					
The quality and extent of help documentation differ throughout the system.					
Moving from one part of a task to another is efficient and user-friendly.					



APPENDIX E
REQUEST LETTER FOR URDANETA POLICE STATION

|CityInsight App

Christian Kaide Cendaña

October 10, 2024

PLTCOL Perlito R. Tuayon Officer in Charge

Urdaneta City Police Station

Brgy. Anonas, Urdaneta City, Pangasinan

Urdaneta City, Pangasinan 2428

Dear PLTCOL Tuayon,

Subject: Request for Crime Indexing Reports for Urdaneta City

We are a group of students from Pangasinan State University, Urdaneta City Campus, currently working on our capstone project titled "**CityInsight: Web and Mobile-based Personal Safety Companion**". As part of our project, we are developing an innovative mobile application aimed at enhancing personal safety in urban environments.

We kindly request that you provide both a printed and soft copy of the following data:

1. Crime indexing reports for the months of June, July, August, and September, including a breakdown per barangay.
2. Crime Indexing Reports for Urdaneta City broken down per barangay within Urdaneta City.
3. Graphs or charts that depict analytics and trends from the crime data.

We respectfully request your support in providing information or allowing us the opportunity to interview you or another knowledgeable member of your team. Your expertise and firsthand experience would offer invaluable insights that closely align with the objectives of our capstone project. We assure you that any information provided will be used solely for academic purposes and will be treated with the utmost confidentiality.

Thank you for your time and consideration of this request. We look forward to your positive response.
Warm regards,

Christian Kaide Cendaña

Group Leader


Christian Kaide Cendaña
Capstone Group Leader

Noted:


NAHUM QUIROS
Capstone Project Adviser


CHRISTINE LOURRINE TABLANTIN, Ph.D
Chairperson IT Department

Approved:


ROY C. FERRER
Campus Executive Director


FREDERICK F. PATACSI
College Dean, College of Computing



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PERSONAL INFORMATION

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Birthplace: Pozorrubio, Pangasinan

Age: 21

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Religion: Roman Catholic

Civil Status: Single

EDUCATION BACKGROUND

Senior High School: **Benigno V. Aldana National High School**

Information And Communications Technology - Strand

Pozorrubio, Pangasinan

2021

Junior High School: **St. Philomena's Academy**

Pozorrubio, Pangasinan

2019

Primary Education: **Imbalbalatong Elementary School**

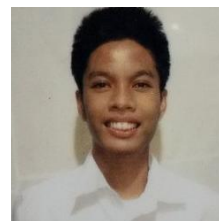
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CURRICULUM VITAE

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Birthplace: San Manuel, Pangasinan

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General Academic Strand

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2021

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Primary Education: **Juan C. Laya Memorial School**

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Senior High School: **Phinma Upang College of Urdaneta**

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2020

Junior High School: **Badipa National High School**

Urdaneta City, Pangasinan

2018

Primary Education: **Badipa Elementary School**

Urdaneta City, Pangasinan

2015

